

Probing Exoplanetary Rings with Asterodensity Profiling: A PhotoRing Analysis of Kepler-51

JORGE I. ZULUAGA,¹ SEBASTIÁN NUMPAQUE,¹ DAVID KIPPING,² AND JAIME A. ALVARADO-MONTES^{3,4,*}

¹SEAP/FACom, Instituto de Física - FCEN, Universidad de Antioquia, Calle 70 No. 52-21, Medellín, Colombia.

²Columbia University, 550 W 120th Street, New York NY 10027

³Australian Astronomical Optics, Macquarie University, Balaclava Road, Sydney, NSW 2109, Australia.

⁴Astrophysics and Space Technologies Research Centre, Macquarie University, Balaclava Road, Sydney, NSW 2109, Australia.

ABSTRACT

The Kepler-51 system hosts transiting super-puff planets whose extremely low bulk densities challenge conventional structural models, suggesting anomalously large inferred radii that might arise from unresolved planetary rings systems. We test this hypothesis by combining asterodensity profiling with a TTV-aware fit of transit light-curves of *Kepler* long-cadence photometry. We find that the transit-inferred stellar densities for planets b and d systematically differ from independent isochrone-based estimates. We discard the photoeccentric effect, demonstrating that the large eccentricities required to explain the density anomalies are robustly ruled out by current dynamical constraints. The density anomalies are also compatible with the **PhotoRing** (PR) effect caused by unmodeled rings. Using a geometric ringed-transit forward model and Bayesian retrieval, we identify configurations that reproduce the transit observables and account for the stellar density anomalies. Our solutions suggest compact, moderately inclined ring systems that increase the planets’ true bulk densities by a factor of 3–4. Rather than ultra-low-density bodies, these objects may be “puff ringed planets,” a scenario consistent with recent JWST observations of Kepler-51d. While the required impact parameters and true stellar densities show tension with independent constraints—which we attribute to Kepler’s 30-minute cadence limiting the resolution of the contact geometry—this study presents the first fully developed application of the PR effect. We provide the complete open-source code to reproduce our results, support further follow-up of Kepler-51, and facilitate applications to other planetary systems.

Keywords: Exoplanet rings (494) — Transit photometry (1709) — Stellar properties (1624) — Monte Carlo methods (2238)

1. INTRODUCTION

Several findings support the hypothesis that planetary rings are not exclusive to our Solar System but a natural outcome of possibly all planetary systems. In the Solar System, all four giant planets possess ring systems (S. Charnoz et al. 2009) linked to tidal disruption of satellites, collisions among moonlets, or residual debris from planetary formation. More recently, the confirmation of rings around small bodies such as Chariklo (F. Braga-Ribas et al. 2014), Quaoar (B. E. Morgado et al. 2023), Chiron (J. L. Ortiz et al. 2023), and the dwarf planet Haumea (J. L. Ortiz et al. 2017) suggests that ring formation can occur under a wide variety of physical conditions, raising more questions about how planetary rings are born and how they evolve. Still, exoplanetary

rings (‘exorings’ for short) remain undiscovered among the thousands of confirmed extrasolar planets. Their detection represents a unique opportunity to explore and provide new constraints on formation and evolution of planetary systems.

Early theoretical investigations demonstrated that Saturn-like ring systems could produce detectable signatures (≤ 100 ppm) in transit photometry (J. W. Barnes & J. J. Fortney 2004), especially during ingress and egress. However, although systematic searches for these signals in *Kepler* and *TESS* data have been conducted over the years (M. Z. Heising et al. 2015; M. Aizawa et al. 2018; T. Umetani et al. 2025), no unambiguous detections of rings around transiting exoplanets have yet been confirmed (T. M. Brown et al. 2001; N. C. Santos et al. 2015; H. P. Osborn et al. 2017; F. J. Ballesteros et al. 2018; A. Lecavelier des Etangs et al. 2017; B. Akinsanmi et al. 2020). In practice, an obscuring structure around

* Macquarie University Research Fellow (MQRF)

1SWASP J140747b is presently considered the only confirmed exoring case (E. E. Mamajek et al. 2012; M. A. Kenworthy & E. E. Mamajek 2015). However, given how young the system is, these “rings” are most likely composed of dust and gas and probably represent the remains of a circumplanetary disc, rather than debris from previous tidal interactions or moon disruptions.

Exorings could also be detected via modern high-contrast imaging techniques (K. B. Follette 2023) or reflected-light photometry (L. Arnold & J. Schneider 2004; M. Sucerquia et al. 2020; J. I. Zuluaga et al. 2022) and polarimetry (A. K. Veenstra et al. 2025). However, because these techniques are still limited, most studies have concentrated on identifying their distinctive imprints in stellar light curves, which change according to their orientation, radial extent, and optical depth (see eg. Y. Ohta et al. 2009; J. I. Zuluaga et al. 2015).

The most salient effect of an exoring on a transit light-curve arises from the fact that the occulting area of a ringless planet can be significantly smaller than that of a ringed planet. If rings are not included in the transit model of the latter, the planetary radius will be overestimated (see e.g., J. I. Zuluaga et al. 2015). This would result in the bulk density being underestimated, which in turn could cause the planet to be incorrectly identified as an unusually low-density, “puffed” planet (G. Bakos et al. 2007). In extreme cases, exorings could explain the anomalous densities of the so-called “super-puffs” or super puff planets ($\rho < 0.1 \text{ g/cm}^3$, see eg. E. J. Lee & E. Chiang 2016), bodies with radii comparable to those of gas giants but only a few times the mass of Earth.

The planetary system Kepler-51 contains some of the most striking examples of so-called super-puff planets. This compact, multi-planet system hosts at least three transiting planets—Kepler-51b, Kepler-51c, and Kepler-51d—with sizes between those of Neptune and Saturn, whose masses have been inferred from transit-timing variation (TTV) analyses (K. Masuda 2014). With estimated masses of 5–6 $M_{\oplus}$, these planets are formally categorized as super-Earths. Their radii, however, are on the order of 6–10 $R_{\oplus}$, comparable to gas giants, which results in extremely low mean densities of $\rho < 0.06 \text{ g cm}^{-3}$. As a consequence, they are currently the least dense—and thus the most extremely inflated—planets known. More recently, a fourth planet, Kepler-51 e, has been suggested (K. Masuda et al. 2024). Its presence is inferred indirectly, and its characteristics are consequently less well determined than those of the other planets in the system.

Several physical processes have been suggested to explain the extremely low densities and inflated radii of the Kepler-51 planets, including extended H-He envelopes,

atmospheric escape and the possibility that they are still at a relatively early evolutionary stage (see, e.g., J. E. Libby-Roberts et al. 2020; L. Wang & F. Dai 2019). Recent JWST observations of Kepler-51d, however, show, on the one hand, no significant photometric signature indicative of an equatorial bulge or a large opaque ring system (C. Lammers & J. N. Winn 2024), and, on the other hand, a sloped transmission spectrum (J. E. Libby-Roberts et al. 2025) that strongly supports the presence of a high-altitude haze layer, rendering the existence of a massive, opaque exoring highly unlikely. While an optically thin, very porous ring around Kepler-51d cannot be entirely excluded, prior theoretical work has already shown that standard rocky rings struggle to account for the extreme sizes of the Kepler-51 planets without assuming exceptionally porous material (A. L. Piro & S. Vissapragada 2020). In turn, the true nature of Kepler-51b remains unclear without further follow-up observations. Owing to this ongoing debate and their peculiar physical characteristics, the Kepler-51 planets serve as an excellent testing ground for the development and implementation of novel methods.

In J. I. Zuluaga et al. (2015) (hereafter ZKSA2015), the so-called **PhotoRing** (PR) effect was introduced, a technique for identifying planetary rings even when transit light curves lack prominent, well-resolved morphological signatures. PR is part of the broader class of astero-density profiling (AP) techniques (D. M. Kipping 2014, hereafter K2014). In AP, the mean stellar density of the host star inferred from transit observables (the “observed density”)—namely the transit duration, depth, impact parameter, and orbital period—is compared with the bulk stellar density obtained from independent stellar characterization methods (the “true density”), such as asteroseismology or stellar isochrone-fitting. When the transit signal is influenced by unmodeled phenomena, such as stellar blending, orbital eccentricity, or the presence of planetary rings, the observed density will differ from the true density.

Despite the potential of the PR effect and the community’s positive reception of the concept, the method has not yet been applied to real planetary systems. With over a decade of accumulated observations, Kepler-51 now emerges as an optimal target to test the method and help unravel the baffling properties of its planets. To this end, we employ here PR to investigate whether the anomalous characteristics of the Kepler-51 planets can at least be partially explained due to the presence of planetary rings. We infer a stellar density from the Kepler transit light curves—among other transit observables—for each planet and compare it with an independent, isochrone-based estimate of the true stellar den-

sity (T. A. Berger et al. 2023), searching for discrepancies indicative of the PR effect. We then applied the geometric forward model for transiting ringed planets of ZKSA2015 to predict the effective transit observables—namely, the transit depth δ and the durations T_{14} and T_{23} —as well as the corresponding $\rho_{\star, \text{obs}}$ for a specified ring configuration.

This paper is organized as follows: In section 2, we introduce the geometric forward model used to predict effective transit observables for ringed planets. section 3 outlines the astrodensity profiling framework and formally describes the PhotoRing effect. In section 4, we present the observational data analysis, including our light-curve fit of the Kepler-51 planets and the derivation of their anomalous transit-inferred stellar densities. section 5 details the Bayesian inference methodology, explicitly describing the likelihood construction and the treatment of nuisance parameters. In section 6, we report the results of our retrieval setups, detailing the inferred ring geometries and evaluating the model through posterior predictive checks. Finally, section 7 provides a discussion of the physical implications of our findings and our concluding remarks.

2. GEOMETRIC MODELING OF RINGED PLANET TRANSITS

Our inference scheme relies on a deterministic, geometry-based forward model that connects a specified ring configuration to the associated effective transit observables. While we follow that relationship via the open-source Python package `exorings`⁵ (ZKSA2015), this work represents an improved and expanded version of such analytical model on planetary ring transits.

The orbital period P , the impact parameter b , and the true stellar density $\rho_{\star, \text{true}}$ are three key input parameters in the particular forward-model implementation adopted in this work. From these quantities, the geometric scaled semi-major axis is obtained via Kepler’s third law (S. Seager & G. Mallén-Ornelas 2003, hereafter SMO2003):

$$\left(\frac{a}{R_{\star}}\right)_{\text{geo}} = \left(\frac{G \rho_{\star, \text{true}} P^2}{3\pi}\right)^{1/3} \quad (1)$$

where G is the gravitational constant. For a circular orbit ($e = 0$), the orbital inclination is then given by

$$\cos i_{\text{orb}} = \frac{b}{(a/R_{\star})_{\text{geo}}} \quad (2)$$

⁵ The original version is available at <https://github.com/facom/exorings>, and an updated release is maintained in this work’s repository at <https://github.com/seap-udea/aPRE>.

Two remarks are worth highlighting. First, by using the stellar density $\rho_{\star, \text{true}}$ as an input parameter instead of the stellar mass $M_{\star}$ and radius $R_{\star}$, we avoid needing prior knowledge of those basic stellar properties for the analysis itself. Second, although we explicitly include $R_{\star}$ when referring to the scaled semi-major axis, all spatial scales are actually expressed in units of the stellar radius. As a result, the absolute value of $R_{\star}$ is not strictly required for the analysis. Still, to interpret the final inferred properties or express them in SI units, we must adopt values for both stellar parameters (or at least for one of them, given that the stellar density is taken to be known).

The ring system is modeled as a geometrically thin annulus extending from an inner radius $R_{\text{in}} = f_i p$ to an outer radius $R_{\text{out}} = f_e p$, where $p \equiv R_p/R_{\star}$ corresponds to the scaled planetary radius and f_i, f_e are dimensionless radii in units of p .

The orientation of the ring relative to the observer is described by two angles defined with respect to the plane of the sky. These angles determine the geometry of the ring’s sky-projected shape, which is modeled as a perfect ellipse with semimajor axis $A = f_e p$ and semiminor axis $B = A \cos i_R$. The orientation angles are i_R , the projected ring inclination (i.e., the dihedral angle between the ring plane and the sky plane, where $i_R = 90^\circ$ represents an edge-on view of the ring), and θ_R , the projected *tilt* angle (the angle between the projected major axis of the ring and the orbital direction, where $\theta_R = 90^\circ$ indicates that the projected major axis is perpendicular to the transit chord). For clarity, Table 1 lists the definitions of all quantities used throughout this study.

Planetary rings are not fully opaque. To describe their opacity, we adopt a wavelength-independent normal attenuation α ⁶. This parameter is then used to derive a blocking factor $\beta = 1 - \alpha^{1/\cos i_R}$ (J. W. Barnes & J. J. Fortney 2004), which incorporates both blocking and absorption effects and scales the ring’s contribution to the total occulting area according to its inclination.

Given these inputs, our forward model computes two classes of observables. First, it evaluates the effective occulting area A_{eff} and the corresponding transit depth:

$$\delta^{\text{geo}} = \frac{A_{\text{eff}}}{\pi R_{\star}^2}. \quad (3)$$

⁶ The normal attenuation α is related to the normal optical depth τ via $\alpha = \exp(-\tau)$. We prefer the parameterization in terms of α because it is strictly bounded between 0 and 1. This improves numerical stability during Bayesian inference and mitigates the pathological behavior of τ in the high-opacity regime ($\tau \gg 1$), where very large variations in optical depth produce only negligible changes in the forward model.

Table 1. Summary of symbols and notation used throughout this work.

Symbol	Description
<i>Nuisance parameters of the forward model</i>	
P	Orbital period [hours]
b	Transit impact parameter [-]
$\rho_{\star, \text{true}}$	Independent stellar density [g cm^{-3}]
<i>Ringed planet parameters of the forward model</i>	
p	$R_p/R_\star$, scaled true planetary radius [-]
f_i, f_e	Inner and outer ring radius [p]
i_R, θ_R	Projected ring inclination and tilt [deg]
α	$\exp -\tau$, Ring normal attenuation [-]
τ	Ring normal opacity [-]
<i>Derived parameters of the forward model</i>	
$(a/R_\star)_{\text{geo}}$	Geometric scaled semi-major axis [-]
i_{orb}	Orbital inclination [deg]
δ^{geo}	Geometric transit depth [ppm]
T_{14}^{geo}	Geometric total transit duration [hours]
T_{23}^{geo}	Geometric full transit duration [hours]
<i>Transit (photometric) observables</i>	
δ	Transit depth [ppm]
T_{14}	Total transit duration [hours]
T_{23}	Full-transit duration [hours]
p_{obs}	Ringless-model scaled planetary radius [-]
<i>Asterodensity profiling derived properties</i>	
$(a/R_\star)_{\text{obs}}$	Scaled semi-major axis [-]
$\rho_{\star, \text{obs}}$	Transit-inferred stellar density [g cm^{-3}]
Ψ	$\rho_{\star, \text{obs}}/\rho_{\star, \text{true}}$, density anomaly [-]
PR	$10 \log_{10} \Psi$, logarithmic PR scale [-]
Θ_{ij}	$(\Psi_i/\Psi_j)^{2/3}$, Ratio of density anomalies [-] (multiplanet asterodensity profiling, MAP)
<i>Bayesian inference</i>	
θ	Ringed planet parameter vector
η	Nuisance parameters
$\mathbf{y}$	Observable vector
$\mathcal{L}$	Likelihood function
$\hat{p}(\mathbf{y})$	KDE estimate of PhotoFIT posterior

In practice, A_{eff} is written as the area of the planetary disk plus the effective projected area of the annulus. The latter is computed using closed-form expressions for the effective projected areas of the outer and inner ring edges

as functions of (f_i, f_e, i_R, α) , including the edge-on limit where only a fraction of the annulus projects outside the planetary disk (Eq. 2 in ZKSA2015). Second, it determines the locations of the contact points along the transit chord. To illustrate the transit geometry with the exact definitions of the contact positions and transit durations, please refer to Figure 10.

The planet-only contacts $x_{p,1\dots4}$ are obtained from the intersection of the planetary disk with the stellar limb at impact parameter b . For the ring contacts $x_{R,1\dots4}$, rather than adopting the approximate analytical intersection from ZKSA2015 that breaks down at certain high inclinations, we employ a robust, exact-like analytical solution based on the Support Function of a convex body (see Appendix A). The four ring-contact positions along the transit chord are given by:

$$\begin{aligned}
 x_{R,1} &\approx -\sqrt{(1+h_L)^2 - b^2} \\
 x_{R,2} &\approx -\sqrt{\max(0, [1-h_L]^2 - b^2)} \\
 x_{R,3} &\approx +\sqrt{\max(0, [1-h_R]^2 - b^2)} \\
 x_{R,4} &\approx +\sqrt{(1+h_R)^2 - b^2},
 \end{aligned} \tag{4}$$

where h_L and h_R are the effective directional radii of the projected ring ellipse at ingress and egress, respectively:

$$\begin{aligned}
 h_L^2 &= A^2(-x_0 \cos \theta_R + b \sin \theta_R)^2 \\
 &\quad + B^2(b \cos \theta_R + x_0 \sin \theta_R)^2 \\
 h_R^2 &= A^2(x_0 \cos \theta_R + b \sin \theta_R)^2 \\
 &\quad + B^2(b \cos \theta_R - x_0 \sin \theta_R)^2
 \end{aligned} \tag{5}$$

(see equations A6 and A5 in the Appendix). These formulae are key to the methodological implementation used here: as they are fully analytical, we can perform computationally intensive numerical procedures without incurring prohibitively long computation times.

The final contact positions are then taken as $x_i = \min[x_{p,i}, x_{R,i}]$, and converted into the ringed transit durations via

$$T_{14}^{\text{geo}} = \frac{P}{2\pi} \arcsin \left[\frac{x_4 - x_1}{(a/R_\star)_{\text{geo}} \sin i_{\text{orb}}} \right] \tag{6}$$

$$T_{23}^{\text{geo}} = \frac{P}{2\pi} \arcsin \left[\frac{x_3 - x_2}{(a/R_\star)_{\text{geo}} \sin i_{\text{orb}}} \right]. \tag{7}$$

This mathematical procedure specifies the forward mapping from the ringed-planet parameters $(p, f_i, f_e, i_R, \theta_R, \alpha)$ to the predicted effective transit observables $(\delta^{\text{geo}}, T_{14}^{\text{geo}}, T_{23}^{\text{geo}})$, under the assumption that the stellar and orbital input parameters P , b , and $\rho_{\star, \text{true}}$ are independently provided.

3. ASTERODENSITY PROFILING

Assuming a spherical, opaque planet with a mass $M_p \ll M_*$, in a circular orbit and transiting a spherical, unblended star of radius R_* with negligible limb darkening at the maximum transit depth position, SMO2003 showed a mean stellar density, $\rho_{*,\text{obs}}$, can be inferred from the transit observables by inverting Equation 1:

$$\rho_{*,\text{obs}} = \frac{3\pi}{GP^2} \left(\frac{a}{R_*} \right)_{\text{obs}}^3. \quad (8)$$

The observed scaled semi-major axis $(a/R_*)_{\text{obs}}$ should not be mistaken for $(a/R_*)_{\text{geo}}$ that is computed in the forward model from the orbital period and the true stellar density (Equation 1). SMO2003 demonstrated that $(a/R_*)_{\text{obs}}$ can be expressed in terms of the total T_{14} and full-transit duration T_{23} , via their Equation 8⁷:

$$\left(\frac{a}{R_*} \right)_{\text{obs}}^2 = \frac{F_{(+)}^2 - b_{\text{obs}}^2 [1 - \sin^2(\pi T_{14}/P)]}{\sin^2(\pi T_{14}/P)}, \quad (9)$$

where $F_{(\pm)} \equiv (1 \pm \sqrt{\delta})$ is an auxiliary variable.

This formula assumes non-grazing transits ($1 < b < 1 - p$), a planet that is much smaller than its host star ($p \ll 1$), and transit durations that are much shorter than the orbital period ($T_{14}/P \ll 1$). These conditions are typically met in most transiting exoplanet systems (SMO2003; K2014), including the Kepler-51 planets.

Under the same assumptions, the observed impact parameter b_{obs} is obtained from the transit shape via Equation 9 in SMO2003:

$$b_{\text{obs}}^2 = \frac{F_{(-)}^2 - \sin^2(\pi T_{23}/P) / \sin^2(\pi T_{14}/P) F_{(+)}^2}{1 - \sin^2(\pi T_{23}/P) / \sin^2(\pi T_{14}/P)}. \quad (10)$$

It is worth noting that equations (9) and (10), and thus Equation 8, are equally valid when using the geometrically computed durations T_{14}^{geo} and T_{23}^{geo} from the forward model, obtained via Equation 6 and Equation 7. The specific context in which the formulae are employed determines which set of quantities should be used. For example, when deriving the observed semi-major axis in the photometric analysis (see subsection 4.2), the time and period values are those inferred from the photometric data, namely δ , T_{14} , and T_{23} . In contrast, during the inference stage we employ instead the geometric quantities δ^{geo} , T_{14}^{geo} , and T_{23}^{geo} (see section 5).

⁷ We distinguish here between the *measured* transit parameters T_{14} and T_{23} and the *calculated* values T_{14}^{geo} and T_{23}^{geo} produced by the forward model

The concept of asterodensity profiling (K2014) is based on the comparison between the photometrically inferred stellar density, $\rho_{*,\text{obs}}$, and an independent estimate of the true stellar density, $\rho_{*,\text{true}}$, obtained through methods such as spectroscopic analysis, asteroseismology, or stellar evolution modeling. If all assumptions of the simple transit model are satisfied, both density estimates should agree within observational uncertainties.

Asterodensity profiling also explores various astrophysical mechanisms for diagnosing violations of the idealized transit assumptions through statistically significant deviations between $\rho_{*,\text{obs}}$ and $\rho_{*,\text{true}}$. One well-known example is the photoeccentric effect, which arises when a planet on an eccentric orbit is incorrectly modeled as circular (K2014). Other mechanisms include contamination by blended companions (photoblend), unocculted stellar activity (photospot), and additional transit-shape distortions such as those induced by planetary rings (photoring, ZKSA2015).

3.1. The PhotoRing Effect

Planetary rings introduce geometric effects that are not captured by the standard spherical-planet transit model. As shown by ZKSA2015, the projected area of a ring system can substantially increase the observed transit depth δ_{obs} . When rings are not included in the transit model, this larger depth δ_{obs} is interpreted as being produced by a larger planetary disk, which leads to an overestimation of the planetary radius R_p .

Rings also modify the contact times of a transit. Because the ring system extends beyond the planetary limb, the first contact occurs earlier and the fourth contact occurs later than in the case of a ringless planet. As a result, the total transit duration T_{14} increases, while the full-transit duration T_{23} becomes shorter. Since these three quantities constitute the fundamental transit observables, these geometric distortions propagate through the transit fitting procedure and bias the inferred values of $(a/R_*)_{\text{obs}}$ and b_{obs} (i.e., equations 9 and 10, respectively). Consequently, the stellar density derived from the transit, $\rho_{*,\text{obs}}$, can be significantly under or overestimated. The resulting systematic offset in the stellar density inferred from transits due to an unaccounted-for ring system was first characterized by ZKSA2015 and is known as the PhotoRing (PR) effect.

By introducing the stellar anomaly ratio, following D. M. Kipping et al. (2012), as

$$\Psi \equiv \left(\frac{\rho_{*,\text{obs}}}{\rho_{*,\text{true}}} \right), \quad (11)$$

the PR effect is expressed using a logarithmic scale that employs the same notation:

$$\text{PR} \equiv 10 \log_{10} \Psi. \quad (12)$$

In Figure 1, we present contours of the PR-effect metric across a wide range of ring orientations. We adopt planetary and stellar parameters akin to those of Kepler-51b, along with tentative properties for the ring system. As shown, PR effect typically causes the stellar density to be underestimated; that is, $\rho_{\star, \text{obs}} < \rho_{\star, \text{true}}$ or $\text{PR} < 0$, with only a relatively small domain where $\rho_{\star, \text{obs}} > \rho_{\star, \text{true}}$. Interestingly, for systems hosting ringed planets—with similar properties to those assumed here—the inferred stellar mean density can be underestimated by as much as a factor of ≈ 2.5 ($\text{PR} \approx -4$) or overestimated by up to a factor of ≈ 1.6 ($\text{PR} \approx +2$).

4. OBSERVATIONAL DATA ANALYSIS

The observational and data foundation of this work rests on posterior distributions of the asteroid density profiling observables ($\delta, \rho_{\star, \text{obs}}, b_{\text{obs}}$) for Kepler-51b and Kepler-51d, the innermost and outermost planets of Kepler-51, obtained from a dedicated light-curve fit of the available *Kepler* photometry (see subsection 4.1). These posteriors are not derived from a direct ring-transit model, they rather represent the inferences made under the standard assumption of a spherical, ringless planet, thus encoding the effective transit observables that must result from any ringed-planet configuration.

In this section, we describe the origin of these posteriors and how they are incorporated into our ring-retrieval framework. Although more recent observations of the system have been obtained with JWST, HST, and TESS facilities (K. Masuda et al. 2024; C. Lammers & J. N. Winn 2024; J. E. Libby-Roberts et al. 2025), the present analysis exclusively relies on constraints derived from the original *Kepler* photometry, without including the aforementioned new datasets.

4.1. Transit and PhotoFIT Modeling

For a robust inference of the transit parameters in the presence of TTVs, without resorting to a full dynamical analysis, we employed a so-called *photometric fit with independent transit times* (hereafter PhotoFIT). Enforcing a strict linear ephemeris on systems with significant TTVs can artificially smear the transit shape. In our PhotoFIT approach, the central transit time (T_c) for each epoch is treated as an independent free parameter during the light curve fitting process. This decouples the transit’s photometric shape from the underlying orbital dynamics, allowing us to accurately measure the geometric transit parameters while also extracting the individual transit epochs.

For Kepler-51b and Kepler-51d, the asteroid density profiling posterior distributions were sampled using the nested sampling algorithm *MultiNest* (F. Feroz et al. 2009, 2019) applied to the *Kepler* photometric time series⁸. The analysis fits each transit simultaneously with our PhotoFIT model, extracting individual mid-transit times for each observed epoch while allowing to estimate the AP parameters from the full ensemble of transits.

For Kepler-51d, the small number of observed transits and the corresponding parameter space allowed a single nested-sampling run, yielding one posterior distribution per parameter. For Kepler-51b, however, the large number of observed epochs translates into a high-dimensional parameter space—since each epoch contributes an individual mid-transit time as a free parameter—making a single nested-sampling run computationally intractable. To handle this high dimensionality, the full photometric time series was divided into three independent temporal segments, each covering a distinct subset of observed transits. A separate run was performed on each segment, producing its corresponding posterior distributions for AP parameters. We then combine the resulting distributions into a single posterior by running a short Markov Chain Monte Carlo sampler that simultaneously draws from the three posterior distributions, treating them as statistically independent constraints on the transit parameters. The resulting chain defines the final posterior of Kepler-51b used throughout this work.

4.2. Derived Observables

For each planet, the main outputs of the PhotoFIT are joint posterior samples of the AP parameters ($\rho_{\star, \text{obs}}, b_{\text{obs}}, \delta$), along with the orbital period P and the limb-darkening coefficients. Assuming the planet has no rings, these parameters capture all of the transit-geometry information encoded in the *Kepler* light curves. The secondary transit quantities $(a/R_{\star})_{\text{obs}}, T_{14}$, and T_{23} , which are needed for our AP analysis, are then obtained by deterministically inverting the relations given in K2014. Their posterior distributions are derived directly from the same posterior samples, ensuring that the full covariance structure inferred from the photometry is preserved. In Figure 2, we show the marginal posterior distributions for the full set of observables ($\rho_{\star, \text{obs}}, \delta, T_{14}, T_{23}, b_{\text{obs}}, (a/R_{\star})_{\text{obs}}$) for both Kepler-51b and Kepler-51d. Our results (shaded curves) are

⁸ We exclude the recently discovered fourth planet, Kepler-51 e, and Kepler-51c due to its grazing transit geometry (K. Masuda et al. 2024)

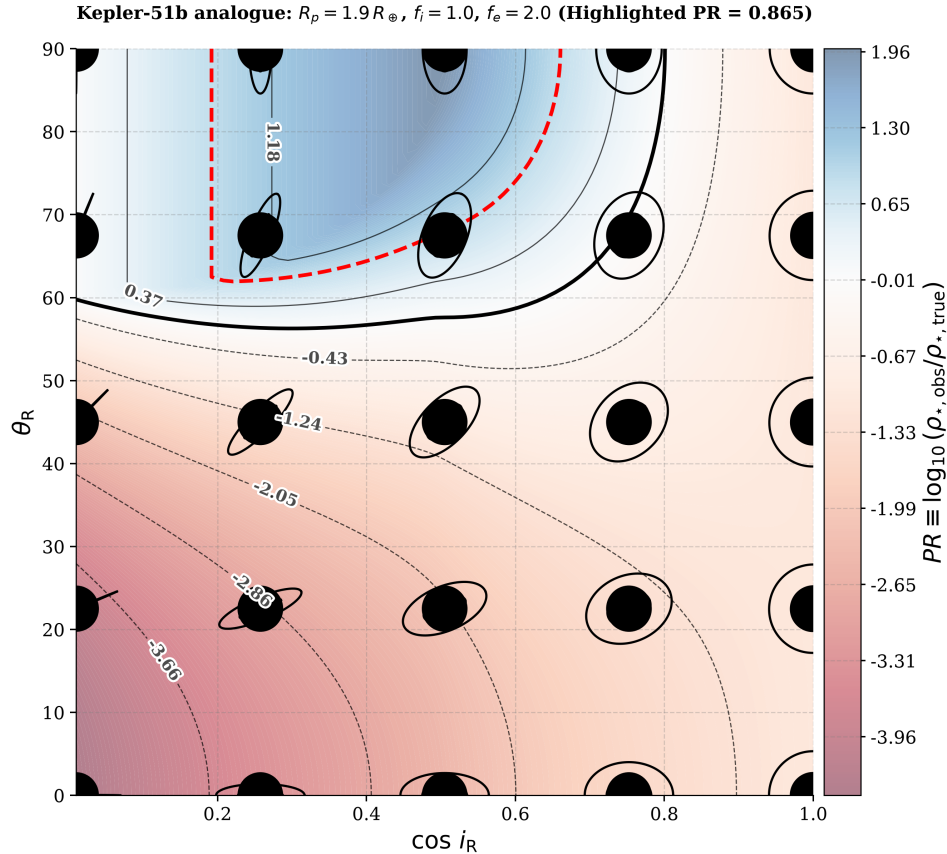


Figure 1. Heat map and contour lines of the PhotoRing (PR) logarithmic scale as a function of the effective ring orientation parameters (i_R, θ_R). PR is evaluated for an analogue of Kepler-51b with a bulk density equal to that of Earth. The thick black contour marks the zero level, $PR = 0$, which identifies configurations that exhibit a density degeneracy (i.e., the inferred stellar density is equal to the true density despite the presence of rings). The highlighted contour correspond to the nominal observed anomaly for the actual planet Kepler-51b according to our PhotoFIT fits (see text for details).

compared with those obtained by other authors (see next section).

As detailed in section 2 and section 3, the geometrical forward model provides, for a specified set of ring parameters ($p, f_i, f_e, i_R, \theta_R, \alpha$) and suitable choices of orbital and stellar properties ($P, b, \rho_{\star, \text{true}}$), a prediction for the corresponding observable vector. The Bayesian inference on the ring parameters is subsequently carried out by confronting this model prediction with the distribution derived from the light curves. This procedure exactly constitutes the foundation of the PR analysis.

4.3. Literature Comparison

The Kepler-51 system has been characterized by several independent photometric analyses spanning a decade: the confirmed three-planet architecture by K. Masuda (2014), the atmospheric characterization of J. E. Libby-Roberts et al. (2020) and J. E. Libby-Roberts et al. (2025), and the refined dynamically re-analysis and discovery of a fourth non-transiting planet, Kepler-51e of K. Masuda et al. (2024). Since our goal

is to obtain posterior distributions of the photometric transit observables that enter the asteroidensity profiling and ring inference described in section 3–section 5, the comparison between our PhotoFIT-derived posteriors and the literature point estimates therefore serves two purposes: (i) to validate that our analysis recovers transit shapes consistent with independent measurements, and (ii) to identify and explain genuine discrepancies that may arise from differences in data selection, dynamical assumptions, or statistical frameworks. In Table 2 we summarize the values of the transit observables derived from our PhotoFIT, along with the corresponding literature values.

A visual comparison of the values reported in the literature with those derived from our photometric analysis is presented in Figure 2. Because the posterior distributions of the observables reported by other authors are not available, we assume in all cases that they follow Gaussian distributions (shown as empty curves). While our own posteriors demonstrate that observable distributions can deviate substantially from normality, this

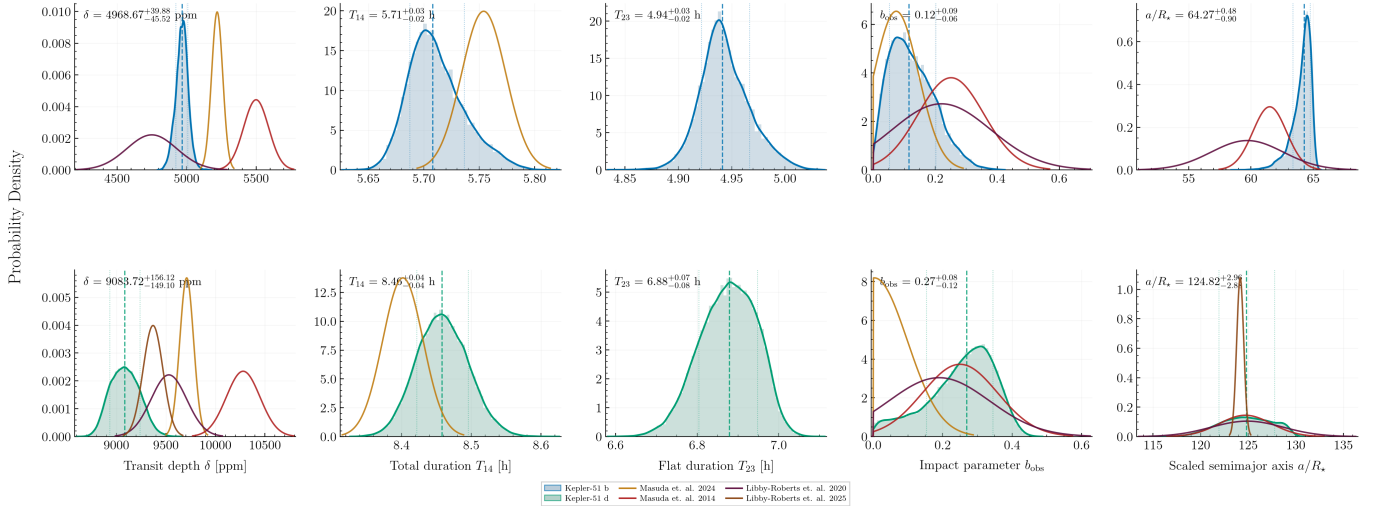


Figure 2. Marginal posterior distributions of the observables for Kepler-51b (top row) and Kepler-51d (bottom row) obtained from the PhotoFIT carried out in this work are shown as filled distributions. Vertical dashed lines and the accompanying labels indicate the median and the 68% credible interval for each marginal posterior. For comparison, literature estimates are depicted as normal distributions (unfilled curves) constructed from the published point values and their reported uncertainties; in the case of K. Masuda et al. (2024), a truncated normal distribution is used for the impact parameter b to respect its physical limits. Only those observables provided by each reference are plotted (for example, none of the alternative sources report values for the flat-bottom duration T_{23}).

assumption is made solely for comparative purposes and the exact shapes of the literature distributions are not used in our inference pipeline.

The transit depth δ is the most straightforward quantity to determine from the data. Table 2 and Figure 2 reveal a pronounced scatter among published values for both planets that is larger than the quoted uncertainties of the individual measurements. The most natural explanation for this spread is *unocculted stellar activity*, rather than any systematic issue in the light-curve fitting. This phenomenon was explicitly detected in the follow-up observations of Kepler-51d using NIRspec, as documented by C. Lammers & J. N. Winn (2024). Kepler-51 is a young (~ 500 Myr), magnetically active star whose evolving starspot distribution alters the apparent out-of-transit flux level, artificially increasing the measured transit depth if the spot contribution is not explicitly modeled. Since the spot covering fraction changes over the *Kepler* observing window, and our analysis does not apply a dedicated correction for stellar spots, the derived depths should be interpreted as an average over the star’s activity state across the different epochs. This explains their agreement with the full span of literature values and supports the conclusion that the observed dispersion is predominantly caused by time-varying stellar activity, rather than by differences in analysis methodology.

On the other hand, the impact parameter b_{obs} is among the least constrained transit observables reported in the literature, with published estimates spanning a wide range for both planets. This behavior arises primarily from the limited temporal resolution of the *Kepler* long-cadence data (~ 30 min), which is comparable to the ingress/egress durations of the planets and therefore averages over and smooths the slopes of these regions that contain most of the information used to constrain the transit chord geometry and, therefore, the impact parameter. This loss of geometric information introduces a strong degeneracy in the determination of b_{obs} that can only be effectively broken with higher-cadence observations, as demonstrated by the JWST analysis of K. Masuda et al. (2024), which resolves a nearly central value for Kepler-51d ($b \sim 0.003$) and also show that, even with high-quality data, the solution for the impact parameter of Kepler-51b remains bimodal, implying the existence of two distinct transit geometries that fit the observations equally well. Our posterior for b_{obs} , obtained from a joint fit of all available epochs, is consistent with both the low- b and high- b solutions reported and shows no significant tension with individual measurements, thereby reflecting the degeneracy imposed by the temporal resolution of the available photometry instead of a physical disagreement.

The total transit duration T_{14} and the flat-bottom duration T_{23} are not frequently tabulated in the literature,

Table 2. Compendium of transit observables for Kepler-51b and d: this work (PhotoFIT posterior) and literature values

Observable	Source	Kepler-51b	Kepler-51d
$p [R_\star (R_\oplus)]$	This work (PhotoFIT)	$0.07055^{+0.00026}_{-0.00030} (6.687^{+0.024}_{-0.028})$	$0.09531^{+0.00082}_{-0.00078} (9.034^{+0.078}_{-0.074})$
	K. Masuda et al. (2024)	$0.07225 \pm 0.0003 (6.83 \pm 0.13)$	$0.09857 \pm 0.00037 (9.32 \pm 0.18)$
	K. Masuda (2014)	$0.07414^{0.00059}_{-0.00061} (7.1 \pm 0.3)$	$0.10141^{0.00084}_{-0.00085} (9.7 \pm 0.5)$
	J. E. Libby-Roberts et al. (2020)	$0.0689 \pm 0.0013 (6.89 \pm 0.14)$	$0.0976 \pm 0.0009 (9.46 \pm 0.16)$
	J. E. Libby-Roberts et al. (2025)	—	$0.09682 \pm 0.00051 (9.19 \pm 0.05)$
δ [ppm]	This work (PhotoFIT)	$4977^{+36.1}_{-42.1}$	9084^{+156}_{-149}
	K. Masuda et al. (2024)	5220 ± 40	9710 ± 70
	K. Masuda (2014)	5500 ± 90	10280 ± 170
	J. E. Libby-Roberts et al. (2020)	4750 ± 180	9530 ± 180
	J. E. Libby-Roberts et al. (2025)	—	9370 ± 100
T_{14} [h]	This work (PhotoFIT)	$5.708^{+0.0238}_{-0.0176}$	$8.458^{+0.0377}_{-0.036}$
	K. Masuda et al. (2024)	5.754 ± 0.02	8.402 ± 0.029
T_{23} [h]	This work (PhotoFIT)	$4.939^{+0.0243}_{-0.019}$	$6.88^{+0.0686}_{-0.0752}$
b_{obs}	This work (PhotoFIT)	$0.1443^{+0.0551}_{-0.0579}$	$0.2696^{+0.0757}_{-0.116}$
	K. Masuda et al. (2024)	0.074 ± 0.072	0.003 ± 0.095
	K. Masuda (2014)	0.251 ± 0.106	0.25 ± 0.108
	J. E. Libby-Roberts et al. (2020)	0.22 ± 0.16	0.19 ± 0.145
$a/R_\star$	This work (PhotoFIT)	$64.08^{+0.46}_{-0.647}$	$124.8^{+2.96}_{-2.88}$
	K. Masuda et al. (2024)	61.5 ± 1.35	124.7 ± 2.75
	J. E. Libby-Roberts et al. (2020)	59.7 ± 2.9	124.9 ± 3.8
	J. E. Libby-Roberts et al. (2025)	—	124.2 ± 0.37

because most works describe the transit shape directly using $(a/R_\star)_{\text{obs}}$ and b_{obs} rather than the contact-time durations themselves. In the one case where T_{14} has been published by K. Masuda et al. (2024), our posterior values for both planets are consistent, differing by at most $\lesssim 4$ minutes, which lies comfortably within the quoted uncertainties. In contrast, the scaled semi-major axis $(a/R_\star)_{\text{obs}}$ is the most consistently determined observable across all datasets and analyses: our measurements for both planets agree with those of K. Masuda et al. (2024), J. E. Libby-Roberts et al. (2020), and the independent JWST-based determination of J. E. Libby-Roberts et al. (2025) to within 1σ . This agreement shows that the orbital scale of the system is effectively unaffected by the cadence, wavelength coverage, or modeling assumptions described above, and that $(a/R_\star)_{\text{obs}}$ is set almost entirely by the precisely measured orbital period and total transit duration.

4.4. Anomalous Stellar Densities

A key consequence of the asterodensity profiling framework is that, within the standard transit model—assuming unblended, spot-free light curves and spherical planets on circular orbits—all transiting planets in the same system must yield an identical value of

$\rho_{\star, \text{obs}}$. This conclusion follows directly from Equation 8, since the inferred stellar density is computed from the measured orbital scale $(a/R_\star)_{\text{obs}}$, which in turn depends only on the transit light-curve morphology and the orbital period. Consequently, any discrepancy in $\rho_{\star, \text{obs}}$ among multiple planets orbiting the same star indicates that the standard transit model is failing to capture some aspect of the actual transit geometry for at least one of the planets, or that the transit light curves are being systematically modified by a physical effect not accounted for in the model.

Figure 3 presents the marginal posterior distributions of $\rho_{\star, \text{obs}}$ derived from our photometric analysis for both planets, together with the corresponding distribution of $\rho_{\star, \text{true}}$ from the independent isochrone-based estimates of T. A. Berger et al. (2023) (shown as filled curves). For comparison, we also display the values reported by J. E. Libby-Roberts et al. (2020) and K. Masuda et al. (2024), modeling their posteriors as normal distributions based on their quoted means and uncertainties. For ease of comparison, the mean values and associated $1\text{-}\sigma$ or 68% uncertainties from our work and from the literature are compiled in Table 3.

Several mechanisms could explain the discrepancy between the stellar density inferred from photometry

Asterodensity Profiling - Kepler 51

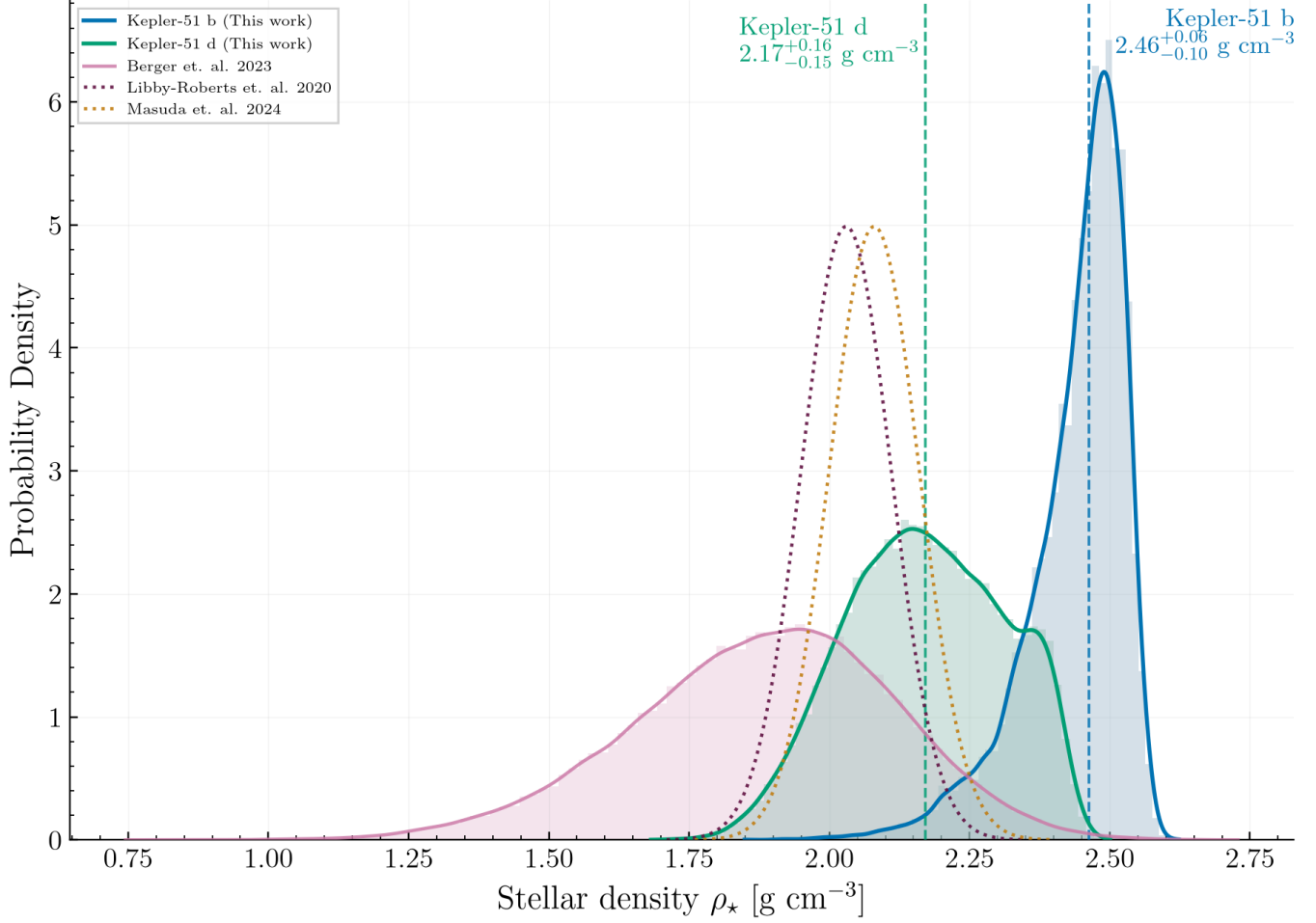


Figure 3. Shown are the marginal posterior distributions of the transit-derived stellar density, $\rho_{\star,\text{obs}}$, for Kepler-51b (blue) and d (orange), obtained from the PhotoFIT presented here. The vertical dashed lines and accompanying labels mark the median and the 68% credible interval for each posterior. The discontinuous curves correspond to Gaussian distributions constructed from literature point estimates and their reported uncertainties from J. E. Libby-Roberts et al. (2020) (purple) and K. Masuda et al. (2024) (green). Also plotted are the independent isochrone-based posterior samples for the true stellar density, $\rho_{\star,\text{true}}$, from T. A. Berger et al. (2023) (filled pink).

and that obtained from independent stellar-astrophysics methods:

- (i) an underlying non-circular orbit, commonly known as the photoeccentric effect (R. I. Dawson & J. A. Johnson 2012; D. M. Kipping et al. 2012; D. M. Kipping 2014). If a planet transits near periastron, its higher orbital velocity mimics a closer orbit, leading to an artificially inflated inferred stellar density.
- (ii) the omission of an explicit spot correction in the photometric analysis, which affects the measured $(a/R_{\star})_{\text{obs}}$ (Equation 9) and, in turn, $\rho_{\star,\text{obs}}$. In our study, this effect could partially explain the

higher stellar densities derived for both planets, as outlined in subsection 4.3.

- (iii) Unmodeled dynamical interactions among the planets. Kepler-51 exhibits large-amplitude transit timing variations (TTVs) that can distort the effective transit light-curve shape if not properly accounted for. In our analysis, we substantially mitigate this bias by fitting a separate mid-transit time for each individual epoch. More generally, an incomplete characterization of the system's architecture can also bias stellar-density estimates. This is demonstrated by the three-planet configuration assumed by K. Masuda (2014), as well as by both the transit-fitting and photodynamical analyses of K. Masuda et al. (2024) (Table 3).

Table 3. Estimates of the stellar density for the Kepler-51 system. The values are grouped according to whether they are obtained from transit observables (including photodynamical modeling) or from independent stellar characterization. The transit-based estimates reported by [K. Masuda et al. \(2024\)](#) were derived under the assumption of the previously known three-planet configuration, whereas their isochrone-based stellar characterization was carried out using the current four-planet model. For reference, we also list an estimate of the logarithmic PR scale ([Equation 12](#)), computed by adopting a true reference stellar density of 2 g cm^{-3} , which is close to the mean of the independent estimates. The uncertainties on PR were computed via straightforward linear error propagation. The PR values are reported for context only and are not used in the inference analysis.

Category	Source	Method	ρ_* [g cm^{-3}]	PR (ref.)
Transit-derived	This work (Kepler-51b)	PhotoFIT posterior	$2.441^{+0.053}_{-0.073}$	0.87 ± 0.12
	This work (Kepler-51d)	PhotoFIT posterior	$2.171^{+0.158}_{-0.147}$	0.36 ± 0.31
	K. Masuda (2014)	Transit light-curve fit	$2.16^{+0.15}_{-0.13}$	0.33 ± 0.28
	K. Masuda et al. (2024)	Transit light-curve fit	2.42 ± 0.06	0.83 ± 0.11
	K. Masuda et al. (2024)	N-body Photodynamical fit	$\sim 2.56^{+0.423}_{-0.34686}$	1.07 ± 0.65
Independent	J. E. Libby-Roberts et al. (2020)	Isochrone fitting (CKS+Gaia DR2)	2.03 ± 0.08	–
	K. Masuda et al. (2024)	Isochrone fitting (MIST+Gaia DR3)	2.08 ± 0.08	–
	T. A. Berger et al. (2023)	Isochrone fitting (Gaia DR3)	$1.896^{+0.218}_{-0.246}$	–

That truncated dynamical model incorrectly predicted the *JWST*-measured mid-transit time of Kepler-51d by ~ 120 minutes. Consequently, stellar-density inferences that rely on this assumption—including those presented here, which likewise neglect perturbations from the fourth planet—may also be affected by systematic biases.

- (iv) The existence of an extra geometric feature—such as a ring system—that changes the observed transit configuration by enlarging the effective occulting area and altering the transit contact times.

4.5. Ruling out the Photoeccentric Effect

The positive density anomaly observed in Kepler-51b and d ($\Psi \equiv \rho_{*,\text{obs}}/\rho_{*,\text{true}} > 1$) could in principle be induced by an unmodeled orbital eccentricity if the planets transit near periastron. This is known as the photoeccentric effect ([R. I. Dawson & J. A. Johnson 2012](#)). To verify whether this mechanism can account for the observed $\rho_{*,\text{obs}}$, we calculate the minimum eccentricity $e_{\min}$ required by the PhotoFIT and compare it against the dynamical constraints imposed by TTVs ([D. M. Kipping 2014](#)).

For a single planet, the minimum eccentricity capable of producing an observed density ratio $\Psi > 1$ (which occurs when the argument of periastron $\omega = 90^\circ$) is given by:

$$e_{\min} = \frac{\Psi^{2/3} - 1}{\Psi^{2/3} + 1}. \quad (13)$$

By evaluating this expression over our posterior samples of Ψ for each planet, we find $e_{\min} = 0.085^{+0.047}_{-0.038}$

for Kepler-51b and $e_{\min} = 0.059^{+0.049}_{-0.037}$ for Kepler-51d. As illustrated in [Figure 4](#), these photometric requirements are severely incompatible with the robust dynamical eccentricities derived from N-body models (e.g. [K. Masuda et al. 2024](#)), which restrict the eccentricities to $e_b \approx 0.016$ and $e_d \approx 0.01 - 0.02$. Explaining the anomalies solely through the photoeccentric effect would require eccentricities roughly 3 to 5 times larger than the allowed dynamical values.

Nevertheless, the eccentricities inferred from the TTV analysis are subject to uncertainties; therefore, there is a small probability that some of these eccentricities yield values of Ψ that are consistent with the PhotoFIT posterior distribution. To quantitatively evaluate this probability, we carried out a Monte Carlo calculation of the expected values of the photoeccentric bias, defined as

$$\Psi^{\text{PE}} \equiv \frac{(1 + e \sin \omega)^3}{(1 - e^2)^{3/2}}.$$

In [Figure 4](#), we present the comparison between the distribution of predicted bias provided the uncertainties in e and $e \sin \omega$ from [K. Masuda \(2014\)](#) and the measured density anomaly for both planets.

As is evident from the predicted PE bias distributions for both planets, the overlap with the observed density anomalies is limited. To formally quantify this, we calculate the probability that the PE effect can fully account for the observed anomaly by determining the fraction of dynamically stable configurations that produce a synthetic density ratio Ψ^{PE} equal to or greater than the observed ratio Ψ . By marginalizing over the dynamical constraints on eccentricity and assuming a

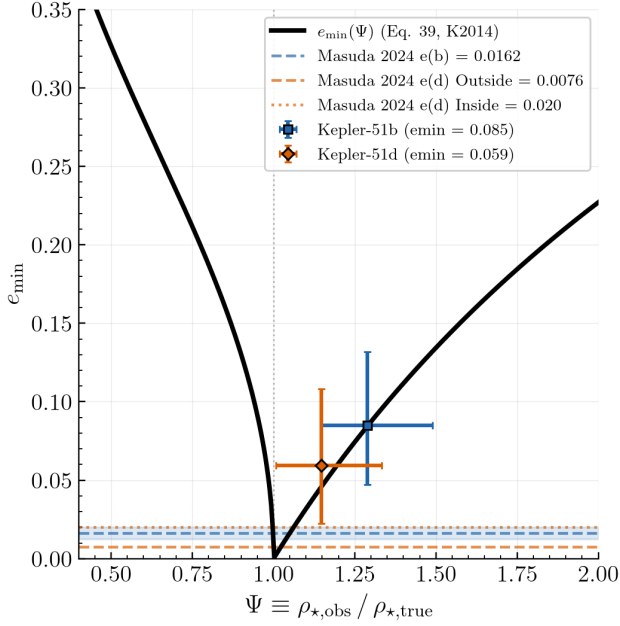


Figure 4. The minimum required eccentricity $e_{\min}$ plotted as a function of the density ratio $\Psi = \rho_{\star, \text{obs}} / \rho_{\star, \text{true}}$ (solid black curve). The dashed curves show the actual eccentricity values obtained from the detailed TTV analysis by K. Masuda et al. (2024).

uniform distribution for the argument of periastron ω , we find that the probability of the PE effect explaining the anomaly is only $\approx 1\%$ for Kepler-51b and $\approx 16\%$ for Kepler-51d. Furthermore, even under the most conservative and optimistic assumption of perfect orbital alignment ($\omega = \pi/2$, yielding the maximum possible PE bias for a given eccentricity), the theoretical upper limits for these probabilities only reach $\approx 3.3\%$ and $\approx 18.7\%$, respectively. These low probabilities strongly disfavor the photoeccentric effect as the primary cause of the anomalously low transit densities.

A final concern remains before we can fully trust the previous result. The analysis relies on the assumption that the true density is known precisely. To conclusively rule out any bias arising from the adopted reference density $\rho_{\star, \text{true}}$, we employ the Multibody Asterodensity Profiling (MAP) framework (D. M. Kipping et al. 2012). MAP exploits the requirement that both planets orbit the same host star. The ratio of their transit-derived densities yields the observable quantity $\Theta_{bd} \equiv (\Psi_b / \Psi_d)^{2/3}$. Using a first-order expansion of the photoeccentric effect, D. M. Kipping et al. (2012) demonstrated that a deviation of Θ_{bd} from unity imposes an absolute lower limit on the combined eccentricity of the pair:

$$(e_b + e_d) \geq \frac{|\Theta_{bd} - 1|}{2} \equiv (e_b + e_d)_{\min}. \quad (14)$$

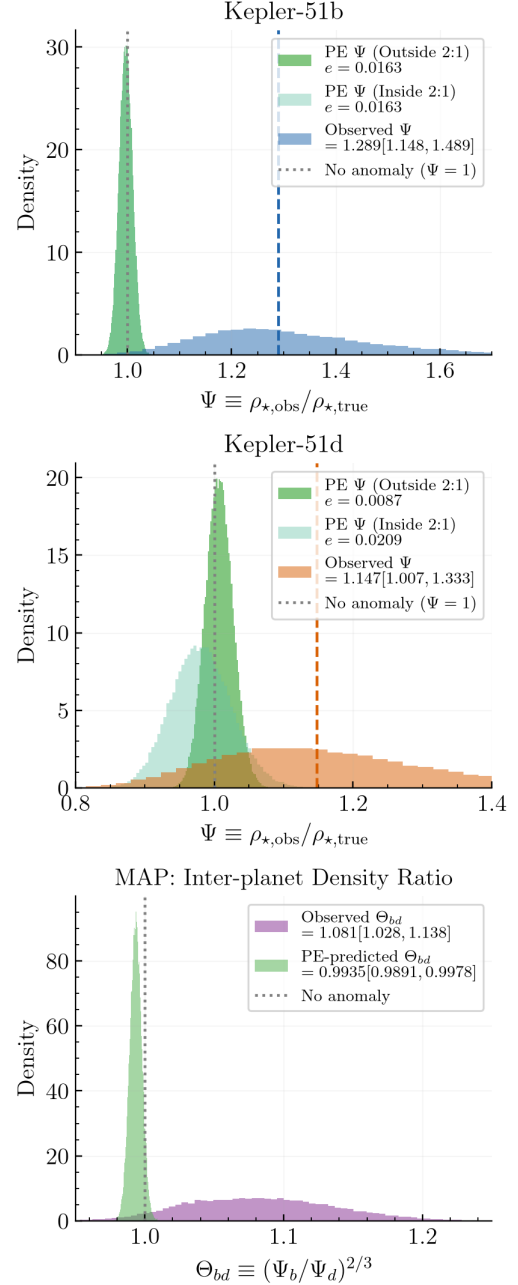


Figure 5. Top panels: Forward prediction of the photoeccentric bias $\Psi(e, \omega)$ versus the observed density anomaly Ψ . Using dynamical posteriors from K. Masuda et al. (2024). Bottom panel: The Multibody Asterodensity Profiling (MAP) analysis which compares the expected inter-planet density ratio $\Theta_{bd} = (\Psi_b / \Psi_d)^{2/3}$ and the observed one.

Our analysis yields an observed ratio $\Psi_b / \Psi_d = 1.125^{+0.090}_{-0.082}$, leading to a MAP-inferred lower bound of $(e_b + e_d)_{\min} \approx 0.041$. However, the sum of the dynamical eccentricities from K. Masuda et al. (2024) is only $(e_b + e_d) \approx 0.024$. The measured eccentricities fall substantially short of the strict multi-planet limit, ruling

out the photoeccentric effect independently of the assumed stellar parameters.

4.6. The *PhotoRing* Hypothesis

Although the discrepancies could reasonably result from scenarios (ii) or (iii), having rigorously ruled out scenario (i) in subsection 4.5, we now turn our attention to scenario (iv). In the remainder of this work we focus on this ringed-planet scenario—not because we consider it the most likely cause of the observed density discrepancies, but in order to evaluate whether a ringed configuration can bring the transit observables into agreement with a self-consistent stellar density. We also emphasize that this is the first study in the literature to perform a full inference using the PR. Therefore, even if other explanations for the density anomalies are possible, using this approach is still a valuable pursuit, as it establishes both a general methodology and computational tools that can be applied to the analysis of other systems.

If the density anomalies are caused by the presence of rings, a notable feature of the specific case under consideration is that the sign of the measured offset in all our analyses falls in the positive-PR regime ($\rho_{\star,\text{obs}} > \rho_{\star,\text{true}}$), which is less frequently encountered than the negative-PR regime expected for most ring orientations (see Figure 1). While this does not rule out the ring hypothesis—since physically plausible configurations include both zero and positive offsets—it does indicate that, should rings be driving the observed anomaly, they must occupy a relatively constrained region of parameter space. In particular, this would require the rings to have relatively large tilt angles, $\theta_R \gtrsim 60^\circ$, with respect to the orbital plane. Such large tilts are physically feasible only if the ring inclination relative to the orbital plane is also substantial. The consequences of these configurations will be examined in a later section.

5. BAYESIAN INFERENCE: FROM OBSERVABLES TO RING PARAMETERS

The forward model introduced in section 2 specifies a deterministic relationship between the ring configuration parameters $\boldsymbol{\theta} = (p, f_i, f_e, i_R, \theta_R, \tau)$ —given a set of nuisance parameters $\boldsymbol{\eta} = (P, \rho_{\star,\text{true}}, b)$ —and the resulting effective transit observables:

$$\boldsymbol{\theta} \mapsto \mathbf{y}(\boldsymbol{\theta}, \boldsymbol{\eta}) \equiv (\rho_{\star,\text{obs}}, \delta^{\text{geo}}, T_{14}^{\text{geo}}, T_{23}^{\text{geo}}, b_{\text{obs}}). \quad (15)$$

Our aim is to identify which ring configurations are compatible with the constraints set by the observed transits, while rigorously accounting for both observational and stellar uncertainties in the Kepler-51 system. To achieve this, we must specify the probability distribution function of the predicted parameters

$(\rho_{\star,\text{obs}}, \delta^{\text{geo}}, T_{14}^{\text{geo}}, T_{23}^{\text{geo}}, b_{\text{obs}})$, conditioned on the available photometric data.

5.1. Likelihood

The most natural choice for the pdf of the transit observables is the joint posterior distribution of the photometric observables $(\rho_{\star,\text{obs}}, \delta, T_{14}, T_{23})$. You may observe that b_{obs} no longer appears in this list, as it also serves as a nuisance parameter. Because the photometric analysis does not yield a continuous distribution, we construct a non-parametric approximation to this function using a multivariate Gaussian kernel density estimator (KDE; D. W. Scott 1992). To do so, we select a statistically representative subsample of the PhotoFIT-derived samples of the observable vector. We denote this subsample by $\{\mathbf{y}_m^{\text{ph}}\}_{m=1}^N$, where each $\mathbf{y}_m$ corresponds to one joint realization of the chosen observables. These samples therefore provide a discrete characterization of the joint posterior distribution of $\mathbf{y}$ implied by the PhotoFIT, which we approximate with the following multivariate Gaussian KDE:

$$\hat{p}(\mathbf{y}) = \frac{1}{N} \sum_{m=1}^N \mathcal{N}(\mathbf{y} | \mathbf{y}_m^{\text{ph}}, \mathbf{H}), \quad (16)$$

where $\mathbf{H} = h^2 \boldsymbol{\Sigma}$ is the KDE bandwidth matrix, constructed from the sample covariance matrix $\boldsymbol{\Sigma}$ and a bandwidth scaling factor h , which determines the degree of smoothing and the correlations of the kernel in the transit observable space. We adopt the standard automated bandwidth selection implemented in the KDE routine, following Scott’s or Silverman’s rule (D. W. Scott 1992; B. W. Silverman 1986).

The resulting continuous function yields an estimate of the probability density across the region of transit observables preferred by the photometric analysis. For any candidate ring configuration $\boldsymbol{\theta}_k$, the forward model generates an associated observable vector, $\mathbf{y}(\boldsymbol{\theta}_k, \boldsymbol{\eta})$ (for the treatment of the nuisance parameters see subsection 5.2 below). We then assess the agreement between this prediction and the photometric posterior by evaluating the KDE density at that point and adopting this value as the likelihood. In this way, correlated uncertainties among the observables are naturally accounted for, without depending on independent point estimates or imposing a specific parametric form for the likelihood:

$$\mathcal{L}(\boldsymbol{\theta}_k) \propto \hat{p}(\mathbf{y} = \mathbf{y}(\boldsymbol{\theta}_k, \boldsymbol{\eta})). \quad (17)$$

The retrieval process consists of sampling the posterior distribution of ring parameters, $\boldsymbol{\theta}$, exploring the region of ring-parameter space whose predicted observables are most consistent with the distribution inferred from the photometric analysis.

5.2. Treatment of Nuisance Parameters

The orbital period P , impact parameter b , and true stellar density $\rho_{\star, \text{true}}$ determine the transit chord geometry and orbital scale via the inferred values of $(a/R_{\star})_{\text{geo}}$ and i_{orb} (Equation 1 and Equation 2). They therefore influence how a given ring configuration translates into the effective transit observables. Because these parameters are not intrinsic to the ring geometry and are instead affected by measurement uncertainties, we treat them as nuisance parameters. Across our retrieval suite, we adopt several different handling strategies for them, ensuring that their uncertainties are propagated into the derived ring properties.

In all setups, P is treated as fixed and set to the literature value (in particular we use the values reported by K. Masuda et al. 2024). Depending on the specific *run configuration*, b and/or $\rho_{\star, \text{true}}$ are either held at fiducial values, explicitly sampled, or effectively marginalized over within the likelihood. When these parameters are sampled, we use a truncated normal prior for the impact parameter, $b \sim \text{TruncNorm}(\mu_b, \sigma_b; [0, 1])$, centered, again, on the ringless-fit value reported by K. Masuda et al. (2024), and a one-dimensional KDE prior for $\rho_{\star, \text{true}}$ based on the isochrone-informed stellar-density posterior from T. A. Berger et al. (2023).

In the most basic setup, when b and/or $\rho_{\star, \text{true}}$ are not treated as sampled parameters, we hold them fixed at their mean values. As a robustness check, we additionally use a pseudo-marginal likelihood in which these parameters are integrated out via Monte Carlo averaging within the likelihood,

$$\mathcal{L}(\theta_{\mathbf{k}}) \approx \frac{1}{M} \sum_{j=1}^M \hat{p}(\mathbf{y} = \mathbf{y}(\theta_{\mathbf{k}}, \boldsymbol{\eta}^{(j)})), \quad (18)$$

where θ represents the remaining ring parameters and the random realizations $\boldsymbol{\eta}^{(j)} = (P, \rho_{\star, \text{true}}^{(j)}, b^{(j)})$ are sampled from their assumed prior distributions. This method incorporates the external uncertainties in b and $\rho_{\star, \text{true}}$ while retaining the original prior information, without embedding these parameters directly into the sampling chain. This corresponds to the proper Bayesian treatment when such parameters are external constraints that should not be updated by the transit data.

5.3. Parameter Space and Priors

In all the cases, the retrieval scheme surveys the parameter space $(f_e, i_R, \theta_R, p, \tau)$, keeping the inner ring radius fixed at $f_i = 1$, because the current photometric precision is insufficient to resolve structures lying inside the ring's outer edge. The planetary radius ratio p is

either allowed to vary freely or held fixed, in order to accommodate the possibility that a portion of the measured transit depth is contributed by the rings themselves. This would imply a smaller planetary radius and could partially account for the unusually low bulk densities inferred for these super-puff planets when no rings are assumed. In the fixed-radius runs, we set $p = p_{\text{min}}$, corresponding to the limiting-density configuration for the planet (see below).

For the planetary and ring parameters, we adopt priors that lie within physically justified intervals. For the ring's outer radius, we adopt a uniform prior $f_e \sim \mathcal{U}(1, f_{e, \text{max}})$. The upper bound $f_{e, \text{max}} = 10$ is chosen based on the stability limit for prograde rings, which can reach roughly ~ 0.5 times the planetary Hill radius (R. C. Domingos et al. 2006). For the ring inclination, we assume an isotropic prior, $p(i_R) \propto \sin i_R$ for $i_R \in [0^\circ, 90^\circ]$, and we adopt a uniform prior for the projected tilt, $\theta_R \sim \mathcal{U}(0^\circ, 90^\circ)$, which spans all possible ring orientations.

For the planetary radius, we assume a uniform prior on p limited by physically motivated bounds, $p_{\text{min}} \leq p \leq p_{\text{obs}}$. Here, p_{obs} is the transit-derived radius under a ringless model (as reported by K. Masuda et al. 2024), and p_{min} is a lower bound set by requiring that the bulk density does not exceed that of a rocky, Earth-like composition, such that $p_{\text{min}} = (R_{\oplus}/R_{\star})(M_p/M_{\oplus})^{1/3}$. We adopt the so-called *Outside 2:1* solution of K. Masuda et al. (2024) (their Table 6), taking $M_p = 6.9 M_{\oplus}$ for both Kepler-51b and Kepler-51d (the reported mass ratios m). With $R_{\star} = 0.869 R_{\odot}$ (T. A. Berger et al. 2023), this yields $p_{\text{min}} = 0.0201$ for both planets (equivalent to $1.90 R_{\oplus}$).

When we allow α to vary freely, we impose a uniform prior $\alpha \sim \mathcal{U}(0, 1)$. For exploratory analyses, we instead fix $\alpha = 0.368$, which represents a ring system with moderate opacity and is similar to the optical depths observed in portions of Saturn's main rings (G. Xystouris et al. 2023). This choice prevents the introduction of an extra parameter that is partially degenerate with the ring size and orientation, thereby lowering the dimensionality and computational expense of the retrieval while still providing a representative reference ring model.

For completeness and to facilitate reproducibility, Table 4 reports the priors adopted in our inference procedure.

5.4. Posterior sampling

We investigate the posterior distribution of the ring parameters using complementary Bayesian sampling techniques. Our main inference framework relies on

Table 4. Prior distributions used in the ring retrieval. The planetary mass and stellar radius employed to calculate $p_{\min}$ are taken from K. Masuda et al. (2024) and T. A. Berger et al. (2023), respectively.

Parameter	Prior	Range
<i>Ring geometry</i>		
f_e	$\mathcal{U}(1, f_{e,\max})$	$[1, 10] R_p$
f_i	Fixed	$f_i = 1$
i_R	$p(i_R) \propto \sin i_R$	$[0^\circ, 90^\circ]$
θ_R	$\mathcal{U}(0^\circ, 90^\circ)$	$[0^\circ, 90^\circ]$
$\alpha^\dagger$	$\mathcal{U}(0, 1)$	$[0, 1]$
<i>Planetary radius[†]</i>		
p	$\mathcal{U}(p_{\min}, p_{\text{obs}})$	$[0.0201, 0.0722] \text{ (b)}$ $[0.0201, 0.0986] \text{ (d)}$
<i>Nuisance parameters^{†a}</i>		
$\rho_{\star, \text{true}}$	T. A. Berger et al. (2023)	$1.896^{+0.218}_{-0.246}$
b	K. Masuda et al. (2024)	$0.074 \pm 0.072 \text{ (b)}$ $0.003 \pm 0.095 \text{ (d)}$
<i>Fixed orbital parameters</i>		
P	Fixed (K. Masuda et al. 2024)	45.154 b 130.186 d

^aParameters indicated with a $\dagger$ are kept fixed in the baseline configurations and are only varied in the fully unconstrained model.

nested sampling implemented via the **dynesty** Python package (J. S. Speagle 2020), which yields both posterior samples and estimates of the Bayesian evidence ($\mathcal{Z}$). The Bayesian evidence, mathematically defined as the integral of the likelihood function over the entire prior volume ($\mathcal{Z} = \int \mathcal{L}(\theta) \pi(\theta) d\theta$), provides a natural metric for model comparison that intrinsically penalizes unnecessarily complex models, often referred to as the Bayesian Occam’s razor. This method is particularly well suited to handling the complex likelihood structure resulting from the KDE-based observational constraints.

We use a run configuration of $N_{\text{live}} = 1200$ for live points, $\Delta \ln Z = 0.01$ as stop condition, and $N_{\text{KDE}} = 5000$ photometric posterior samples to train the likelihood. To validate the robustness of the inferred posterior structure, we additionally analyze selected configurations using the affine-invariant ensemble MCMC sampler **emcee** (D. Foreman-Mackey et al. 2013). The agreement between both methods provides an independent consistency attention on ring parameter constraints and exploration mode.

6. INFERRED RING GEOMETRIES

The problem we face is highly dimensional: 6 dimensions correspond to the ringed planet parameters ($p, f_i, f_e, i_R, \theta_R, \tau$), plus 3 nuisance parameters ($P, \rho_{\star, \text{true}}, b$), while the likelihood-KDE is defined in an additional 4-dimensional space ($\rho_{\star, \text{obs}}, \delta, T_{14}, T_{23}$). With so many degrees of freedom and the added complexity of the numerical KDE, the fitting procedure may be subject to significant numerical and mathematical limitations. Our goal is to reduce this complexity while avoiding the loss of essential features of the data and the information they contain.

The minimal dimensionality reduction occurs when, as described in subsection 5.2, we treat the nuisance parameter P (the orbital period) as constant and fix it to the value reported in the literature, which leaves only 2 remaining nuisance parameters. In addition, in all configurations we impose $f_i = 1$, because the current photometric precision is too low to resolve any substructure at the outer boundary of the ring. This choice further reduces the dimensionality of the parameter space to 5. Under these assumptions, the model takes as input parameters $\theta = (f_e, i_R, \theta_R, p, \tau)$, treats $\eta = (P, \rho_{\star, \text{true}}, b)$ as nuisance parameters, and considers $\mathbf{y} = (\rho_{\star, \text{obs}}, \delta, T_{14}, T_{23})$ as observables or KDE variables. We refer to this specific choice of input, nuisance, and observable parameters as the *full-set retrieval setup*.

In Figure 6 and Figure 7, we present the outcomes of fitting the data for both planets using this full-set configuration. At first glance, two notable features emerge from the schematic representation of the planet in the corner plot and the median values reported in the figure title (summarized also in Table 5). First, the planetary depiction is consistent with our expectation that, since PR₀ for both planets, the favored ring configurations correspond to highly inclined rings located in the upper part of Figure 1. Second, we find that the resulting solutions yield planetary radii comparable to those inferred from the ring-free analysis, indicating that the ringed model likewise requires inflated planets, now encircled by rings. We call these solutions “puff ringed planets”.

Another remarkable aspect of the fit is the tension between the values of the true density when it is treated as a nuisance parameter (filled area in the marginal plot at the second row from bottom to top) and the independent estimate of this same quantity (dashed curve). An analogous tension appears for the nuisance parameter b . Since these two parameters set the orbital scale and the transit chord, respectively, their combined freedom enables the forward model to trade off the ring-contact geometry against the transit durations, without artificially forcing the ring parameters to absorb a geometric

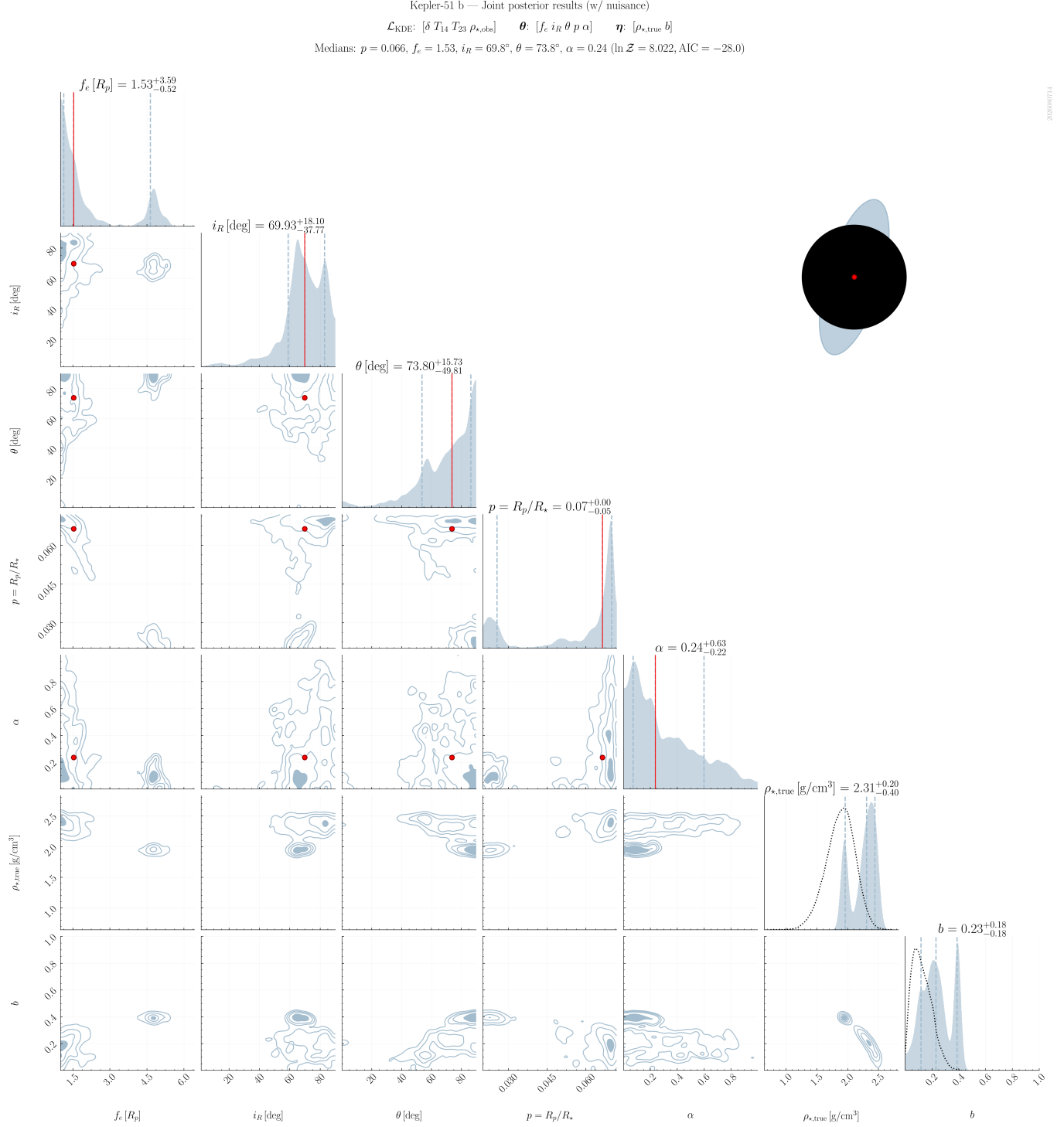


Figure 6. Posterior joint distribution for the PR inference of Kepler-51b for a full-set retrieval setup. The corner-plot panels display 2D and 1D marginal posteriors for both the ringed planet parameters and the nuisance parameters. Together with the marginal posterior distribution of the nuisance parameters, we also show the observed pdf for both parameters (dashed lines). For visualization, we include a schematic of the ring system constructed from the median values of the marginal posteriors, indicated by red lines on the 1D panels. The dark disk denotes the planet, while the filled elliptical annulus represents the ring. Red dots on the panels indicate where the ringed planet parameter is positioned that matches the schematic illustration.

Table 5. Full-set retrieval results for Kepler-51b and Kepler-51d. The table reports the median values and the 16th and 84th percentiles for the ring geometry parameters, nuisance parameters, and selected derived physical properties. We assume a stellar radius of $0.869 R_{\odot}$ (T. A. Berger et al. 2023) and a planetary mass of $6.9 M_{\oplus}$ for both planets (K. Masuda et al. 2024). For comparison, under a ringless model, the inferred planetary radii are $6.85 R_{\oplus}$ (Kepler-51b) and $9.36 R_{\oplus}$ (Kepler-51d), yielding anomalous densities of 0.11 g cm^{-3} and 0.038 g cm^{-3} , respectively.

Parameter	Kepler-51b	Kepler-51d
<i>Ring geometry</i>		
$f_e [R_p]$	$1.53^{+3.11}_{-0.40}$	$1.84^{+5.25}_{-0.56}$
$i_R [^{\circ}]$	$69.8^{+13.1}_{-11.0}$	$63.7^{+15.0}_{-21.8}$
$\theta_R [^{\circ}]$	$73.8^{+12.7}_{-20.2}$	$71.1^{+11.8}_{-20.2}$
p	$0.0664^{+0.0036}_{-0.0408}$	$0.0721^{+0.0210}_{-0.0392}$
α	$0.236^{+0.361}_{-0.165}$	$0.473^{+0.325}_{-0.312}$
<i>Nuisance parameters</i>		
$\rho_{\star, \text{true}} [\text{g cm}^{-3}]$	$2.31^{+0.14}_{-0.35}$	$2.16^{+0.20}_{-0.18}$
b	$0.229^{+0.157}_{-0.111}$	$0.271^{+0.084}_{-0.147}$
<i>Derived parameters</i>		
$R_p [R_{\oplus}]$	6.30	6.84
$\rho_p [\text{g cm}^{-3}]$	0.152	0.119

discrepancy they cannot physically account for. Nevertheless, the sampling procedure fails to obtain solutions fully consistent with the observed data when the assumed true stellar densities are too low. Notably, this tension could, in principle, be exploited to constrain the true stellar density with this method.

6.1. Predictive Posterior Checks

Finding a solution to the retrieval problem is not, by itself, sufficient to ensure that the inferred model actually explains the observed transit features—most importantly the measured stellar density, which is ultimately what we aim to account for. To assess whether the nested samples produced by our retrievals can recover the marginal posterior distributions of the key observables from the photometric analysis we conduct goodness-of-fit evaluations using posterior predictive checks (PPCs).

Concretely, for each retrieval configuration, we propagate the corresponding posterior samples through the forward model presented in section 2. We then compare the resulting marginal posterior distributions of the transit observables with those derived directly from the photometric data (section 4). For additional validation, we also consider observables that were not explicitly in-

cluded in the likelihood function. The agreement for each observable is quantified using two statistics (see, e.g., A. Ramdas et al. 2017): (i) the 1-Wasserstein distance W_1 , which measures the mean absolute displacement between the two empirical distributions; and (ii) the energy distance E , which measures the squared discrepancy between the two distributions.

In Figure 8, we present and compare the marginal posterior distributions (displayed as histograms) of the observables ($\rho_{\star, \text{obs}}, \delta, T_{14}, T_{23}$) derived from the photometric analysis with those obtained from the full-set retrieval configuration, whose selected outcomes are shown in Figure 6 and Figure 7. For both planets, we confirm that the ring model reproduces all observed light-curve properties with high fidelity, a conclusion further supported by the corresponding goodness-of-fit metrics.

6.2. Grid search

After inspecting the solution obtained from the full-set retrieval, several characteristics of the parameter space in which the problem is defined become apparent. First, for both planets we identify regions of parameter space where one or two values of the planetary radius (third row of the corner plots) stand out with higher probabilities than the rest.

For Kepler-51b, for instance, we find a radius close to the minimum allowed value (first peak in the marginal distribution of the third row of Figure 6) which, when compared with the marginal posterior in the p - f_e plane (third row), corresponds to a large ring with $f_e \approx 5$ (second peak in the upper marginal posterior) and a relatively modest opacity (maximum in the α - p panel also in the third row). The alternative “solution” corresponds to a planet whose radius is nearly identical to that obtained from the ringless model (second peak in the marginal of the third row). As anticipated, in this configuration the ring radius can be very small (first peak in the upper marginal), and the opacity is essentially unconstrained (islands in the α - p marginal of the third row). An analogous analysis can be performed for Kepler-51d, based on an examination of the panels in Figure 6 and the insights obtained from the marginal distributions.

This situation indicates that it could be interesting to focus our search on limited regions of parameter space where promising solutions tend to cluster, rather than presuming that the median values of the properties correspond to the best fit. To investigate these specific regions, we performed a grid search. For this exploration, we performed retrievals where the planetary radius p and τ were fixed on a grid composed of 5 values of p , evenly distributed between the minimum value $p_{\min}$

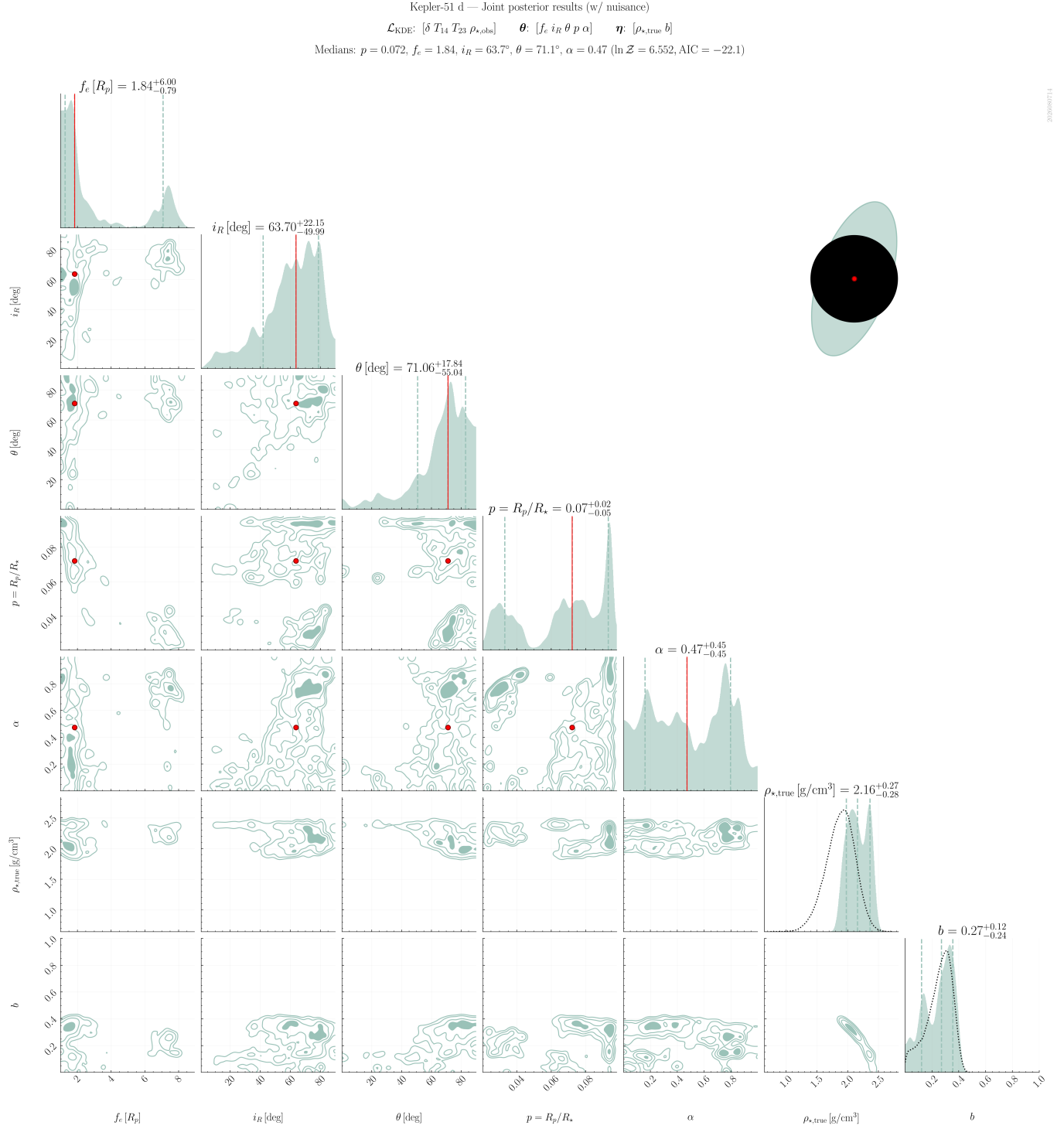


Figure 7. Posterior joint distribution for the PR inference of Kepler-51d for the full-set retrieval setup (see text). The figure specifications are identical to those in Figure 6.

and the observed value p_{obs} (excluding both endpoints), and three values of α (0.2, 0.5, 0.8). The selection of the grid resolution and its specific values was somewhat arbitrary, and a more exhaustive investigation could be undertaken. Our aim was to probe these regions to identify potential local likelihood maxima.

In Table 6 we present the best retrieval results in the grid search for both planets. The retrievals shown are those that fulfill a set of conditions evaluating the quality of the fit, including excellent posterior predictive checks (with normalized deviations $z_1 \lesssim 2$ and $z_{\text{crit}} \lesssim 2$ in the transit observables), a tight agreement with the independently derived true stellar density ($\Delta\rho_{\star, \text{true}} \lesssim 0.3 \text{ g cm}^{-3}$), strongly peaked posteriors for the ring inclination, and a lower limit on the ring outer radius indicating a physically distinct structure ($f_e > 1.1$ at the 16th percentile). We refer to this set of optimal solutions as the “Golden Sample”.

As shown in Table 6, several of the local configurations with fixed p and α yield a higher Bayesian evidence ($\ln \mathcal{Z}$) than the corresponding full-set retrieval where these parameters were left free. This behavior is a direct consequence of the Occam’s razor effect inherent to the Bayesian evidence, which integrates the likelihood over the entire prior volume. In the highly-degenerate full-set retrieval, the sampler must explore a much larger prior space, including regions of low likelihood, which penalizes the total evidence. By fixing p and α near a local likelihood peak, the prior volume is drastically reduced in those dimensions, effectively avoiding the volume penalty. Additionally, breaking the degeneracies between the ring geometry and the planetary radius improves the numerical efficiency of the Nested Sampling algorithm, allowing it to better resolve the local probability maxima.

7. DISCUSSION AND CONCLUSIONS

The framework developed in this work was designed as an exploratory tool to assess whether ring configurations exist that are consistent with the available Kepler-51 transit data, limited also by the ~ 30 min instrumental cadence. Our Bayesian retrievals recover ring configurations capable of reproducing the full set of PhotoFIT-derived transit observables for both planets. However, these solutions require both $\rho_{\star, \text{true}}$ and b inconsistent with their current best external constraints from T. A. Berger et al. (2023) and K. Masuda et al. (2024), preventing a fully self-consistent ring solution from *Kepler* photometry alone. This reflects the intrinsic degeneracy of the problem: numerical convergence toward compatible ring geometries does not guarantee physical viability, which ultimately depends on

tighter independent constraints. It is important to emphasize that our analysis does not constitute a direct fit of a ringed-planet transit model to the *Kepler* light curves. Instead, it is based on the global transit observables set $(\rho_{\star, \text{obs}}, \delta, T_{14}, T_{23}, b_{\text{obs}}, (a/R_{\star})_{\text{obs}})$, which capture only part of the information contained in the full transit morphology. A direct fit to the photometry, ideally performed jointly with a dynamical TTV model, would exploit the detailed ingress and egress shape and any residual signatures to further constrain the ring geometry and break some of the remaining parameter degeneracies in our analysis.

Although we do not obtain a fully self-consistent ring solution, our preferred retrievals for both planets converge toward ring systems with variable extensions ($f_e \approx 1.3\text{--}6.2 R_p$ for Kepler-51b and $f_e \approx 1.3\text{--}3.9 R_p$ for Kepler-51d), moderately inclined ring planes ($i_R \approx 50^\circ\text{--}80^\circ$), and similarly moderate projected tilts ($\theta_R \approx 60^\circ\text{--}80^\circ$). Among the explored solutions, we favor this configuration over either nearly edge-on or face-on geometries, as it is also more consistent with current theoretical expectations for ring formation and long-term stability. This interpretation further suggests a potential detectability bias. As illustrated by our forward model and the PhotoRing effect map (Figure 1), the strongest photometric signatures arise from large, optically thick rings observed at high obliquity. The configurations preferred by our retrieval instead occupy a region of parameter space where the expected photometric signatures are comparatively weaker. If similar ring geometries are common among exoplanets, current searches would be intrinsically less sensitive to them, suggesting that at least part of the apparent lack of confirmed exoplanetary rings may arise from observational selection effects rather than a genuinely low occurrence rate.

The anomalous bulk densities under the ring hypothesis neither exclude alternative explanations for the inflated radii of the Kepler-51 planets nor are they mutually exclusive with them. Atmospheric inflation mechanisms such as high-altitude hazes, primordial H/He envelopes, or a young evolutionary state remain viable scenarios (J. E. Libby-Roberts et al. 2020; L. Wang & F. Dai 2019; C. Lammers & J. N. Winn 2024). Similarly, while the flat transmission spectra observed with HST and *JWST* are commonly attributed to aerosols that mute molecular features (J. E. Libby-Roberts et al. 2025), an optically thick ring system could produce the same effect by attenuating stellar light achromatically. Distinguishing between these interpretations would require detecting wavelength-dependent ring scattering in reflected light, potentially including polarimetric signatures as predicted by A. K. Veenstra et al. (2025). More

Table 6. Golden Sample retrievals from the radius-alpha grid search for Kepler-51b and Kepler-51d.

Planet	Input selection							Other	Metrics
	$p [R_\star] (R_\oplus)$	$f_e [R_p]$	$i_R [^\circ]$	$\theta_R [^\circ]$	α	$\rho_{\star, \text{true}} [\text{g cm}^{-3}]$	b	$\rho_p [\text{g cm}^{-3}]$	$\ln \mathcal{Z}$
Kepler-51b	0.0253 (2.40)	$6.20^{+1.64}_{-2.52}$	$78.4^{+4.3}_{-26.0}$	$79.1^{+8.2}_{-18.5}$	0.20	$2.18^{+0.23}_{-0.08}$	$0.30^{+0.04}_{-0.15}$	2.754	8.49
	0.0670 (6.36)	$1.37^{+0.27}_{-0.13}$	$68.5^{+13.6}_{-22.1}$	$73.6^{+11.2}_{-15.1}$	0.50	$2.30^{+0.08}_{-0.10}$	$0.24^{+0.05}_{-0.05}$	0.148	8.34
	0.0670 (6.36)	$1.38^{+0.46}_{-0.16}$	$72.0^{+13.1}_{-20.5}$	$72.3^{+12.4}_{-16.8}$	0.20	$2.30^{+0.08}_{-0.09}$	$0.23^{+0.05}_{-0.05}$	0.148	8.12
	0.0357 (3.39)	$3.15^{+1.07}_{-0.58}$	$66.6^{+10.9}_{-15.1}$	$69.7^{+10.4}_{-7.3}$	0.20	$2.17^{+0.15}_{-0.13}$	$0.31^{+0.06}_{-0.08}$	0.976	8.09
	0.0670 (6.36)	$1.55^{+0.19}_{-0.10}$	$70.0^{+12.1}_{-15.8}$	$77.3^{+8.9}_{-12.0}$	0.80	$2.30^{+0.07}_{-0.09}$	$0.23^{+0.04}_{-0.05}$	0.148	8.05
	0.0357 (3.39)	$3.63^{+1.07}_{-0.60}$	$70.9^{+9.0}_{-14.2}$	$71.0^{+10.2}_{-6.4}$	0.50	$2.25^{+0.19}_{-0.14}$	$0.27^{+0.07}_{-0.15}$	0.976	7.84
	0.0566 (5.37)	$1.84^{+0.62}_{-0.25}$	$59.7^{+19.8}_{-18.7}$	$74.7^{+7.6}_{-11.6}$	0.50	$2.31^{+0.09}_{-0.17}$	$0.23^{+0.14}_{-0.11}$	0.245	7.74
	0.0462 (4.38)	$2.10^{+0.55}_{-0.28}$	$57.1^{+13.9}_{-15.9}$	$66.9^{+13.5}_{-7.9}$	0.20	$2.22^{+0.14}_{-0.13}$	$0.28^{+0.06}_{-0.08}$	0.452	7.42
	0.0462 (4.38)	$3.39^{+0.42}_{-0.19}$	$71.6^{+7.4}_{-5.5}$	$81.5^{+6.0}_{-6.3}$	0.80	$2.25^{+0.12}_{-0.09}$	$0.26^{+0.04}_{-0.07}$	0.452	7.33
	0.0462 (4.38)	$2.32^{+0.38}_{-0.17}$	$58.3^{+11.5}_{-9.6}$	$78.1^{+8.7}_{-10.8}$	0.50	$2.22^{+0.16}_{-0.15}$	$0.28^{+0.07}_{-0.10}$	0.452	7.08
	0.0566 (5.37)	$2.66^{+0.74}_{-0.29}$	$77.3^{+8.1}_{-11.5}$	$77.7^{+7.0}_{-7.6}$	0.80	$2.31^{+0.11}_{-0.16}$	$0.23^{+0.08}_{-0.08}$	0.245	6.99
Kepler-51d	0.0907 (8.61)	$1.49^{+1.06}_{-0.26}$	$78.5^{+9.7}_{-25.1}$	$69.4^{+13.4}_{-24.3}$	0.20	$2.22^{+0.12}_{-0.18}$	$0.23^{+0.09}_{-0.07}$	0.060	6.73
	0.0907 (8.61)	$1.36^{+0.64}_{-0.17}$	$68.3^{+18.3}_{-32.5}$	$64.4^{+17.5}_{-24.3}$	0.50	$2.22^{+0.12}_{-0.13}$	$0.24^{+0.07}_{-0.08}$	0.060	6.60
	0.0436 (4.14)	$3.31^{+2.83}_{-0.77}$	$63.5^{+19.4}_{-31.6}$	$62.8^{+11.9}_{-12.0}$	0.20	$2.21^{+0.13}_{-0.15}$	$0.25^{+0.08}_{-0.08}$	0.536	6.59
	0.0907 (8.61)	$1.51^{+0.24}_{-0.13}$	$66.1^{+17.2}_{-20.4}$	$71.0^{+13.1}_{-16.1}$	0.80	$2.23^{+0.13}_{-0.15}$	$0.23^{+0.08}_{-0.10}$	0.060	6.33
	0.0436 (4.14)	$3.92^{+1.77}_{-0.64}$	$69.6^{+12.0}_{-16.5}$	$70.1^{+9.2}_{-7.5}$	0.50	$2.19^{+0.15}_{-0.13}$	$0.26^{+0.06}_{-0.11}$	0.536	5.72
	0.0593 (5.63)	$2.11^{+0.72}_{-0.29}$	$52.7^{+18.7}_{-22.6}$	$61.3^{+10.0}_{-11.1}$	0.20	$2.17^{+0.15}_{-0.15}$	$0.27^{+0.07}_{-0.10}$	0.213	5.47
	0.0593 (5.63)	$3.61^{+0.56}_{-0.25}$	$71.6^{+8.6}_{-7.9}$	$78.5^{+6.9}_{-7.5}$	0.80	$2.21^{+0.12}_{-0.11}$	$0.25^{+0.06}_{-0.08}$	0.213	5.36
	0.0750 (7.12)	$2.62^{+0.44}_{-0.26}$	$72.7^{+9.6}_{-10.9}$	$78.3^{+7.6}_{-8.5}$	0.80	$2.17^{+0.11}_{-0.16}$	$0.27^{+0.07}_{-0.07}$	0.105	4.83

generally, a planet hosting both an extended atmosphere and an inclined ring system would appear doubly inflated, with both effects contributing to the observed transit depth. As revealed by our grid search, the inferred bulk densities can increase by modest factors of ~ 2 –5 in some local probability maxima, which still leaves the planets substantially less dense than typical sub-Neptunes. However, we also find equally valid configurations with significantly smaller planetary radii where the bulk density increases by factors of ~ 15 up to ~ 100 , placing them well within the typical density range of volatile-rich or even rocky planets. This multimodal density landscape suggests that rings, if present, can either provide a partial or a full explanation for the super-puff phenomenon depending on the exact system geometry.

An important advantage of asterodensity profiling is its computational efficiency, as deriving $\rho_{\star, \text{obs}}$ requires only the fundamental transit observables ($\delta, T_{14}, T_{23}, P$). This makes the *PhotoRing* diagnostic readily applicable to the full population of super-puff planets discovered by *Kepler* and *K2* without dedicated ring-transit modeling. While multi-planet architectures offer a distinct advantage by allowing the cross-testing of inter-planetary inconsistencies in $\rho_{\star, \text{obs}}$, single-planet systems require a far more exhaustive analysis due to the lack of baseline references for contrast. Nevertheless, any system displaying such inter-planetary discrepancies, similar to

those identified here, remains a prime target for detailed ring retrieval analyses. The case of HIP-41378f already provides the closest precedent, as its super-puff-like density was reconciled with an inclined ring system capable of inflating the apparent transit radius (B. Akin-sanmi et al. 2020). If *PhotoRing* offsets are found to be common among super-puffs, this would support the idea that unresolved ring systems contribute systematically to their apparent radius inflation, with important implications for sub-Neptune occurrence rates and for the theoretical models invoked to explain their formation.

More broadly, the inference framework developed here, which combines *PhotoFIT*-based asterodensity profiling, a geometric ringed-planet transit model, and Bayesian retrieval, is readily transferable to other systems requiring only posterior distributions for the transit observables and an independent estimate of $\rho_{\star, \text{true}}$. Its main limitation is the photometric cadence required to resolve ingress and egress morphology. In the near term, existing *JWST* observations offers the most promising opportunity to test this scenario, as *JWST* can directly constrain b from resolved contact slopes while its spectroscopic capabilities, together with those of *ARIEL*, can simultaneously probing its atmospheric properties. The upcoming high-precision observations from *PLATO* (F. Matuszewski et al. 2023) for thousands of cold Jupiters will also enable systematic searches for ring signatures, improving TTV-based mass constraints,

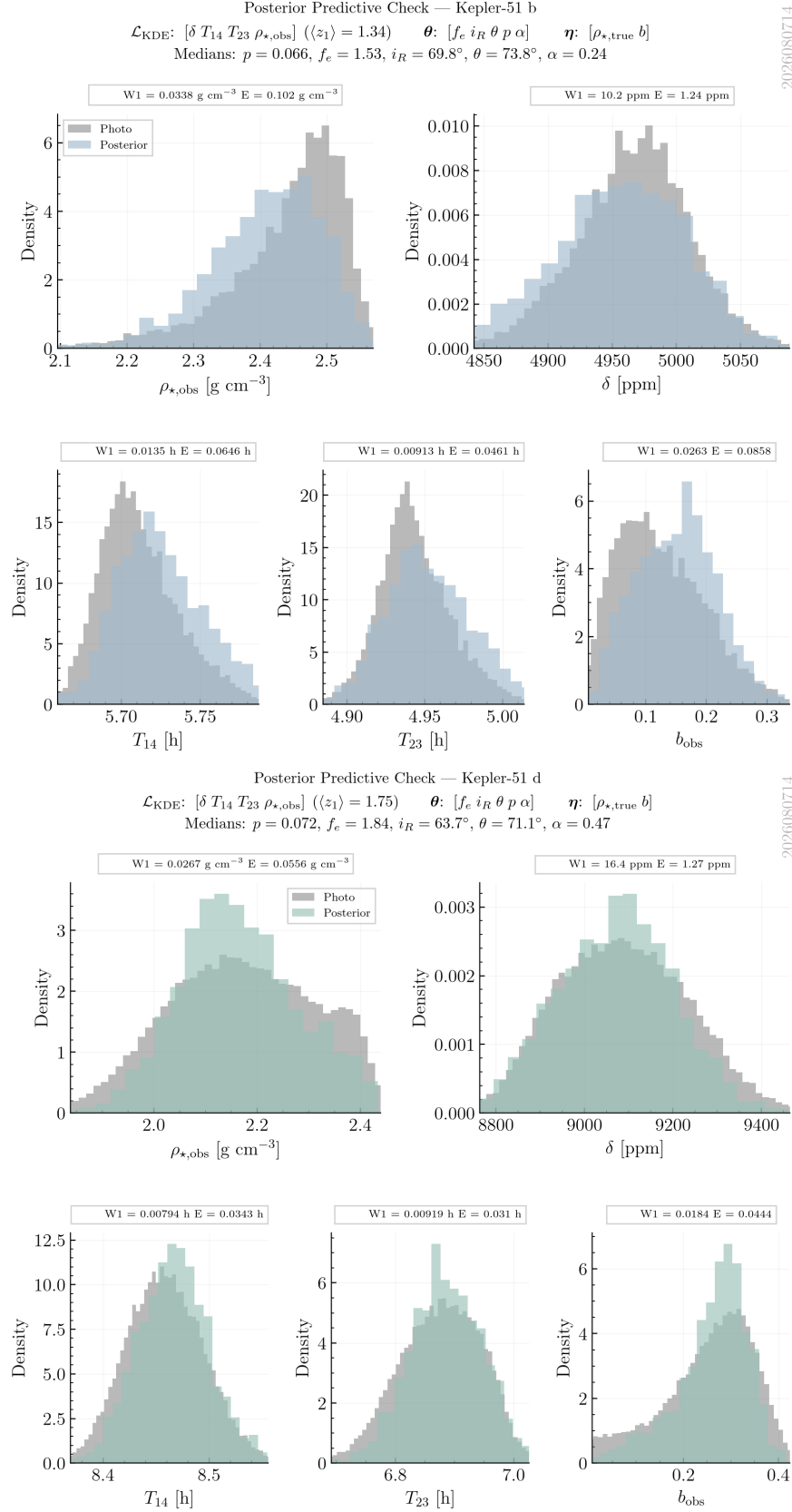


Figure 8. Posterior predictive checks (PPCs) for Kepler-51b and Kepler-51d in the full-set retrieval setup. PPCs compare the model-predicted distribution (blue) with the photometrically derived posterior (grey) for each transit observable. The reported values of W_1 and E (see text) quantify the self-consistent goodness-of-fit between the ringed-planet model and the observed transit observables.

and identifying planets with anomalously low ringless densities.

DATA AVAILABILITY

All the data necessary to reproduce the results and recreate the figures in this work—including the Jupyter notebooks used for model training and evaluation—are publicly accessible in the GitHub repository <https://github.com/seap-udea/PRisma> (PhotoRing inference and modeling algorithm). We anticipate that this project will provide a foundation for future implementations of the PR inference pipeline proposed in this paper. Contributions from the community are encouraged.

ACKNOWLEDGMENTS

This work has been possible due to the availability of open-source, general-purpose Python packages, including Matplotlib (J. D. Hunter 2007), Numpy (C. R. Harris et al. 2020), Scipy (P. Virtanen et al. 2020), mpi4py (L. Dalcin & Y.-L. L. Fang 2021) and pandas (W. McKinney et al. 2010), dynesty (J. S. Speagle 2020) and emcee (D. Foreman-Mackey et al. 2013).

APPENDIX

A. ANALYTICAL DERIVATION OF RING CONTACT POSITIONS USING THE SUPPORT FUNCTION

A.1. Error Analysis of the 2015 Analytical Formula

In the original formulation of the PhotoRing effect by J. I. Zuluaga et al. (2015), calculating the exact temporal contact points between a stellar limb and a projected exoplanetary ring requires solving a non-linear system of trigonometric equations. The previous analytical approximation assumed a highly elongated ellipse ($A \gg B$) and small contact angles, leading to geometrical artifacts at high projected tilts ($\theta_R \gtrsim 60^\circ$) and high inclinations.

To illustrate this, Figure 9 compares the PhotoRing effect contours generated by the precise semi-analytical model **geotrans** with those produced by the original **exorings** analytical approximation from J. I. Zuluaga et al. (2015). As shown, the original formulation breaks down in certain inclination ranges, creating unphysical artifacts.

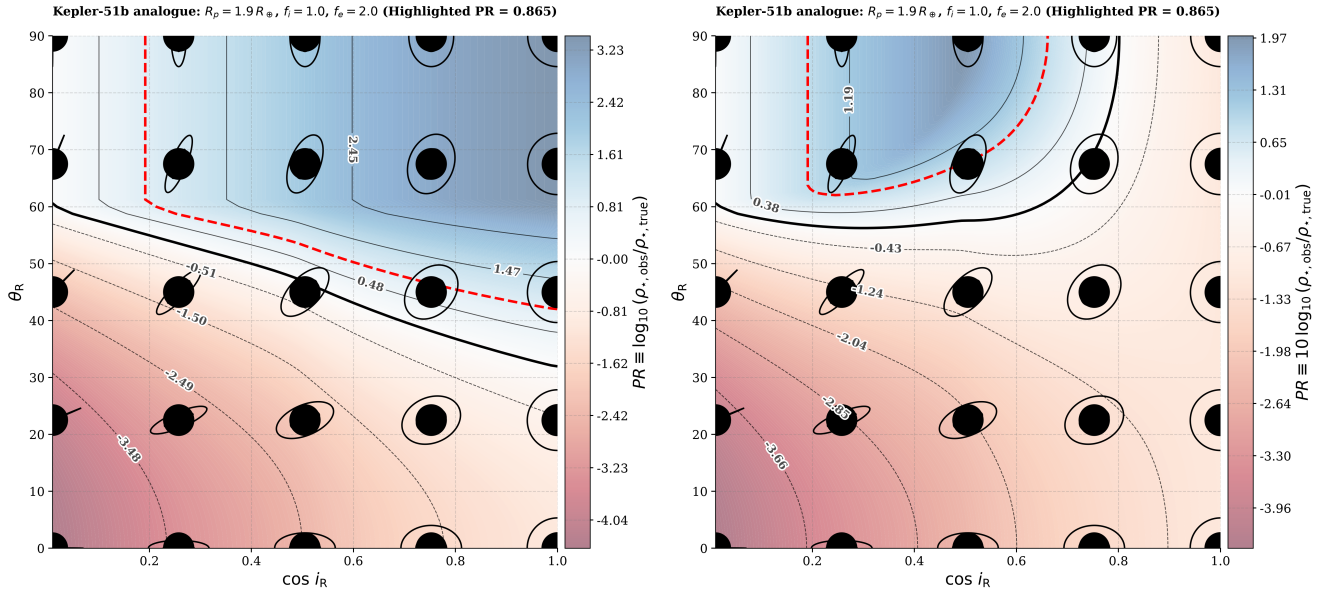


Figure 9. Comparison of the PhotoRing effect contours for Kepler-51b using the original **exorings** analytical approximation from J. I. Zuluaga et al. (2015) (left) and the precise semi-analytical model **geotrans** (right). The original model exhibits geometrical artifacts at certain high projected tilts and inclinations. You can compare the panel on the right with Figure 1, which was obtained using the refined analytical model presented in this appendix.

To provide a robust, exact-like analytical solution suitable for computationally intensive sampling (e.g., Nested Sampling), we reframe the contact geometry using the Minkowski Addition and the Support Function of a convex body (R. Schneider 1993).

A.2. Theoretical Framework: The Support Function

Geometrically, the contact between the stellar disk (a circle of radius $R_* = 1$) and the projected ring (an ellipse with semi-axes A, B rotated by θ_R) occurs when the distance between their centers equals the sum of their effective radii in the direction of contact.

For a convex shape, the distance from its center to a tangent line perpendicular to a given normal vector $\vec{n}$ is defined by its Support Function, $h(\vec{n})$. For an ellipse aligned with the coordinate axes, the support function is:

$$h(\vec{n}) = \sqrt{A^2 n_x^2 + B^2 n_y^2} \quad (\text{A1})$$

A graphical representation of these geometric quantities and the contact positions used in this derivation is provided in Figure 10.

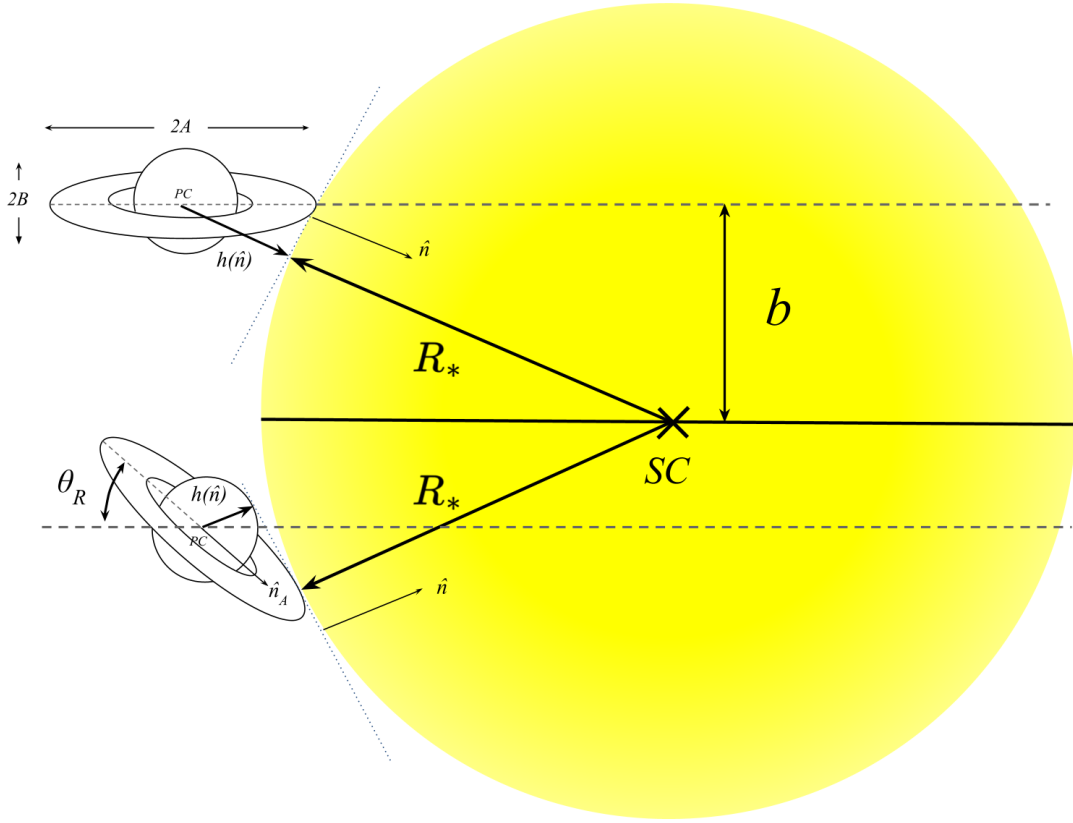


Figure 10. Illustration of the quantities involved in the analytical model for two configurations: one where the rings are oriented coinciding with the coordinate system (left/top), and another where the rings are oriented with an inclination and projected tilt (right/bottom).

When the ellipse is rotated by an angle θ_R (the projected ring tilt), the normal vector must be rotated into the ellipse's local frame. The unit vectors defining the major and minor axes are:

$$\hat{u}_A = (\cos \theta_R, \sin \theta_R), \quad \hat{u}_B = (-\sin \theta_R, \cos \theta_R)$$

The directional effective radius of the rotated ring is then:

$$h(\vec{n}) = \sqrt{A^2 (\vec{n} \cdot \hat{u}_A)^2 + B^2 (\vec{n} \cdot \hat{u}_B)^2} \quad (\text{A2})$$

A.3. The Unperturbed Normal Approximation

To avoid solving for the exact contact normal iteratively, we make a highly accurate physical approximation: the normal vector at the point of contact is nearly identical to the normal vector if the planet were a spherical point-mass.

For a point-mass transit with an impact parameter b , the contact points on the stellar limb have the coordinates $y = b$ and $x = \pm\sqrt{1 - b^2}$. Therefore, the normal vectors at the egress (trailing, R) and ingress (leading, L) contacts are simply:

$$\vec{n}_R = \left(+\sqrt{1 - b^2}, b \right) \quad (\text{A3})$$

$$\vec{n}_L = \left(-\sqrt{1 - b^2}, b \right) \quad (\text{A4})$$

Let $x_0 = \sqrt{1 - b^2}$. We can now evaluate the Support Function of the ring in these two specific directions to obtain the effective contact radii, h_R and h_L .

Projecting $\vec{n}_R$ onto the ellipse axes yields the egress effective radius:

$$\vec{n}_R \cdot \hat{u}_A = x_0 \cos \theta_R + b \sin \theta_R$$

$$\vec{n}_R \cdot \hat{u}_B = -x_0 \sin \theta_R + b \cos \theta_R$$

Squaring and substituting into the support function gives:

$$\begin{aligned} h_R^2 &= A^2(x_0 \cos \theta_R + b \sin \theta_R)^2 \\ &\quad + B^2(b \cos \theta_R - x_0 \sin \theta_R)^2 \end{aligned} \quad (\text{A5})$$

Similarly, for the ingress effective radius (h_L):

$$\vec{n}_L \cdot \hat{u}_A = -x_0 \cos \theta_R + b \sin \theta_R$$

$$\vec{n}_L \cdot \hat{u}_B = x_0 \sin \theta_R + b \cos \theta_R$$

Yielding:

$$\begin{aligned} h_L^2 &= A^2(-x_0 \cos \theta_R + b \sin \theta_R)^2 \\ &\quad + B^2(b \cos \theta_R + x_0 \sin \theta_R)^2 \end{aligned} \quad (\text{A6})$$

A.4. Final Contact Equations

With the effective geometrical radii h_R and h_L defined, the complex ring behaves instantaneously like a sphere of radius $h_{L,R}$. We substitute these directional radii into the standard spherical contact equations. The four contact positions along the x -axis are therefore given exactly by:

Ingress Contacts (Left):

$$x_{R,1} \approx -\sqrt{(1 + h_L)^2 - b^2} \quad (\text{First contact, external}) \quad (\text{A7})$$

$$x_{R,2} \approx -\sqrt{\max(0, [1 - h_L]^2 - b^2)} \quad (\text{Second contact, internal}) \quad (\text{A8})$$

Egress Contacts (Right):

$$x_{R,3} \approx +\sqrt{\max(0, [1 - h_R]^2 - b^2)} \quad (\text{Third contact, internal}) \quad (\text{A9})$$

$$x_{R,4} \approx +\sqrt{(1 + h_R)^2 - b^2} \quad (\text{Fourth contact, external}) \quad (\text{A10})$$

The argument of the square root for internal contacts ($x_{R,2}, x_{R,3}$) is bounded to ≥ 0 to properly handle grazing configurations where a full interior transit does not occur.

REFERENCES

- | | |
|--|--|
| <p>Aizawa, M., Masuda, K., Kawahara, H., & Suto, Y. 2018, AJ, 155, 206, doi: 10.3847/1538-3881/aab9a1</p> | <p>Akinsanmi, B., Santos, N. C., Faria, J. P., et al. 2020, A&A, 635, L8, doi: 10.1051/0004-6361/202037618</p> |
|--|--|

- Arnold, L., & Schneider, J. 2004, *A&A*, 420, 1153,
doi: [10.1051/0004-6361:20035720](https://doi.org/10.1051/0004-6361:20035720)
- Bakos, G., Noyes, R., Kovács, G., et al. 2007, *The Astrophysical Journal*, 656, 552
- Ballesteros, F. J., Arnalte-Mur, P., Fernandez-Soto, A., & Martínez, V. J. 2018, *MNRAS*, 473, L21,
doi: [10.1093/mnrasl/slx105](https://doi.org/10.1093/mnrasl/slx105)
- Barnes, J. W., & Fortney, J. J. 2004, *ApJ*, 616, 1193,
doi: [10.1086/425067](https://doi.org/10.1086/425067)
- Berger, T. A., Schlieder, J. E., & Huber, D. 2023, arXiv e-prints, arXiv:2301.11338,
doi: [10.48550/arXiv.2301.11338](https://doi.org/10.48550/arXiv.2301.11338)
- Braga-Ribas, F., Sicardy, B., Ortiz, J. L., et al. 2014, *Nature*, 508, 72, doi: [10.1038/nature13155](https://doi.org/10.1038/nature13155)
- Brown, T. M., Charbonneau, D., Gilliland, R. L., Noyes, R. W., & Burrows, A. 2001, *ApJ*, 552, 699,
doi: [10.1086/320580](https://doi.org/10.1086/320580)
- Charnoz, S., Dones, L., Esposito, L. W., Estrada, P. R., & Hedman, M. M. 2009, in *Saturn from Cassini-Huygens*, ed. M. K. Dougherty, L. W. Esposito, & S. M. Krimigis, 537, doi: [10.1007/978-1-4020-9217-6_17](https://doi.org/10.1007/978-1-4020-9217-6_17)
- Dalcin, L., & Fang, Y.-L. L. 2021, *Computing in Science & Engineering*, 23, 47
- Dawson, R. I., & Johnson, J. A. 2012, *ApJ*, 756, 122,
doi: [10.1088/0004-637X/756/2/122](https://doi.org/10.1088/0004-637X/756/2/122)
- Domingos, R. C., Winter, O. C., & Yokoyama, T. 2006, *MNRAS*, 373, 1227,
doi: [10.1111/j.1365-2966.2006.11104.x](https://doi.org/10.1111/j.1365-2966.2006.11104.x)
- Feroz, F., Hobson, M. P., & Bridges, M. 2009, *Monthly Notices of the Royal Astronomical Society*, 398, 1601,
doi: [10.1111/j.1365-2966.2009.14548.x](https://doi.org/10.1111/j.1365-2966.2009.14548.x)
- Feroz, F., Hobson, M. P., Cameron, E., & Pettitt, A. N. 2019, *The Open Journal of Astrophysics*, 2, 10,
doi: [10.21105/astro.1306.2144](https://doi.org/10.21105/astro.1306.2144)
- Follette, K. B. 2023, *PASP*, 135, 093001,
doi: [10.1088/1538-3873/aceb31](https://doi.org/10.1088/1538-3873/aceb31)
- Foreman-Mackey, D., Hogg, D. W., Lang, D., & Goodman, J. 2013, *PASP*, 125, 306, doi: [10.1086/670067](https://doi.org/10.1086/670067)
- Harris, C. R., Millman, K. J., van der Walt, S. J., et al. 2020, *Nature*, 585, 357, doi: [10.1038/s41586-020-2649-2](https://doi.org/10.1038/s41586-020-2649-2)
- Heising, M. Z., Marcy, G. W., & Schlichting, H. E. 2015, *ApJ*, 814, 81, doi: [10.1088/0004-637X/814/1/81](https://doi.org/10.1088/0004-637X/814/1/81)
- Hunter, J. D. 2007, *Computing in Science and Engineering*, 9, 90, doi: [10.1109/MCSE.2007.55](https://doi.org/10.1109/MCSE.2007.55)
- Kenworthy, M. A., & Mamajek, E. E. 2015, *ApJ*, 800, 126,
doi: [10.1088/0004-637X/800/2/126](https://doi.org/10.1088/0004-637X/800/2/126)
- Kipping, D. M. 2014, *MNRAS*, 440, 2164,
doi: [10.1093/mnras/stu318](https://doi.org/10.1093/mnras/stu318)
- Kipping, D. M., Dunn, W. R., Jasinski, J. M., & Manthri, V. P. 2012, *Monthly Notices of the Royal Astronomical Society*, 421, 1166
- Lammers, C., & Winn, J. N. 2024, *ApJL*, 977, L1,
doi: [10.3847/2041-8213/ad91ae](https://doi.org/10.3847/2041-8213/ad91ae)
- Lecavelier des Etangs, A., Hébrard, G., Blandin, S., et al. 2017, *A&A*, 603, A115,
doi: [10.1051/0004-6361/201730554](https://doi.org/10.1051/0004-6361/201730554)
- Lee, E. J., & Chiang, E. 2016, *ApJ*, 817, 90,
doi: [10.3847/0004-637X/817/2/90](https://doi.org/10.3847/0004-637X/817/2/90)
- Libby-Roberts, J. E., Berta-Thompson, Z. K., Désert, J.-M., et al. 2020, *AJ*, 159, 57,
doi: [10.3847/1538-3881/ab5d36](https://doi.org/10.3847/1538-3881/ab5d36)
- Libby-Roberts, J. E., Bello-Arufe, A., Berta-Thompson, Z. K., et al. 2025, arXiv e-prints, arXiv:2505.21358,
doi: [10.48550/arXiv.2505.21358](https://doi.org/10.48550/arXiv.2505.21358)
- Mamajek, E. E., Quillen, A. C., Pecaute, M. J., et al. 2012, *AJ*, 143, 72, doi: [10.1088/0004-6256/143/3/72](https://doi.org/10.1088/0004-6256/143/3/72)
- Masuda, K. 2014, *ApJ*, 783, 53,
doi: [10.1088/0004-637X/783/1/53](https://doi.org/10.1088/0004-637X/783/1/53)
- Masuda, K., Libby-Roberts, J. E., Livingston, J. H., et al. 2024, *AJ*, 168, 294, doi: [10.3847/1538-3881/ad83d3](https://doi.org/10.3847/1538-3881/ad83d3)
- Matuszewski, F., Nettelmann, N., Cabrera, J., Börner, A., & Rauer, H. 2023, *A&A*, 677, A133,
doi: [10.1051/0004-6361/202245287](https://doi.org/10.1051/0004-6361/202245287)
- McKinney, W., et al. 2010, *scipy*, 445, 51
- Morgado, B. E., Sicardy, B., Braga-Ribas, F., et al. 2023, *Nature*, 614, 239, doi: [10.1038/s41586-022-05629-6](https://doi.org/10.1038/s41586-022-05629-6)
- Ohta, Y., Taruya, A., & Suto, Y. 2009, *ApJ*, 690, 1,
doi: [10.1088/0004-637X/690/1/1](https://doi.org/10.1088/0004-637X/690/1/1)
- Ortiz, J. L., Santos-Sanz, P., Sicardy, B., et al. 2017, *Nature*, 550, 219, doi: [10.1038/nature24051](https://doi.org/10.1038/nature24051)
- Ortiz, J. L., Pereira, C. L., Sicardy, B., et al. 2023, *A&A*, 676, L12, doi: [10.1051/0004-6361/202347025](https://doi.org/10.1051/0004-6361/202347025)
- Osborn, H. P., Rodriguez, J. E., Kenworthy, M. A., et al. 2017, *MNRAS*, 471, 740, doi: [10.1093/mnras/stx1249](https://doi.org/10.1093/mnras/stx1249)
- Piro, A. L., & Vissapragada, S. 2020, *The Astronomical Journal*, 159, 131
- Ramdas, A., Trillos, N. G., & Cuturi, M. 2017, *Entropy*, 19, 47, doi: [10.3390/e19020047](https://doi.org/10.3390/e19020047)
- Santos, N. C., Martins, J. H. C., Boué, G., et al. 2015, *A&A*, 583, A50, doi: [10.1051/0004-6361/201526673](https://doi.org/10.1051/0004-6361/201526673)
- Schneider, R. 1993, *Convex bodies: the Brunn-Minkowski theory*, Vol. 151 (Cambridge university press Cambridge)
- Scott, D. W. 1992, *Multivariate Density Estimation: Theory, Practice, and Visualization* (New York: John Wiley & Sons)
- Seager, S., & Mallén-Ornelas, G. 2003, *ApJ*, 585, 1038,
doi: [10.1086/346105](https://doi.org/10.1086/346105)

- 1426 Silverman, B. W. 1986, *Monographs on Statistics and*
 1427 *Applied Probability*, Vol. 26, Density Estimation for
 1428 *Statistics and Data Analysis* (London: Chapman and
 1429 Hall)
- 1430 Speagle, J. S. 2020, *MNRAS*, 493, 3132,
 1431 doi: [10.1093/mnras/staa278](https://doi.org/10.1093/mnras/staa278)
- 1432 Sucerquia, M., Alvarado-Montes, J. A., Zuluaga, J. I.,
 1433 Montesinos, M., & Bayo, A. 2020, *MNRAS*, 496, L85,
 1434 doi: [10.1093/mnrasl/slaa080](https://doi.org/10.1093/mnrasl/slaa080)
- 1435 Umetani, T., Aizawa, M., Ezoe, Y., & Ishisaki, Y. 2025,
 1436 *Publications of the Astronomical Society of Japan*, 77, 86
- 1437 Veenstra, A. K., Zuluaga, J. I., Alvarado-Montes, J. A.,
 1438 Sucerquia, M., & Stam, D. M. 2025, *A&A*, 693, A310,
 1439 doi: [10.1051/0004-6361/202347194](https://doi.org/10.1051/0004-6361/202347194)
- 1440 Virtanen, P., Gommers, R., Oliphant, T. E., et al. 2020,
 1441 *Nature Methods*, 17, 261, doi: [10.1038/s41592-019-0686-2](https://doi.org/10.1038/s41592-019-0686-2)
- 1442 Wang, L., & Dai, F. 2019, *ApJL*, 873, L1,
 1443 doi: [10.3847/2041-8213/ab0653](https://doi.org/10.3847/2041-8213/ab0653)
- 1444 Xystouris, G., Arridge, C. S., Morooka, M. M., & Wahlund,
 1445 J.-E. 2023, *Monthly Notices of the Royal Astronomical*
 1446 *Society*, 526, 5839, doi: [10.1093/mnras/stad2793](https://doi.org/10.1093/mnras/stad2793)
- 1447 Zuluaga, J. I., Kipping, D. M., Sucerquia, M., & Alvarado,
 1448 J. A. 2015, *ApJL*, 803, L14,
 1449 doi: [10.1088/2041-8205/803/1/L14](https://doi.org/10.1088/2041-8205/803/1/L14)
- 1450 Zuluaga, J. I., Sucerquia, M., & Alvarado-Montes, J. A.
 1451 2022, *Astronomy and Computing*, 40, 100623,
 1452 doi: [10.1016/j.ascom.2022.100623](https://doi.org/10.1016/j.ascom.2022.100623)